

Online Bayesian Inversion for an Anchored Multi-Tag Visual–Inertial Rig

Foundations, inference structure, and practical implementation

Geometry, priors, likelihoods, estimator hierarchy, and prior-sampling illustrations

Abstract

This report develops an event-driven Bayesian treatment of trajectory and pose estimation for a rigid body carrying a camera and an IMU in an anchored multi-tag environment. One fiducial tag fixes the world frame, while the remaining tags are stationary but initially unknown. The reference model is a posterior on a continuous trajectory together with a static tag map and visual nuisance terms, conditioned on an asynchronous stream of IMU packets and camera frames processed in chronological order. IMU packets contribute inertial transition likelihoods; camera frames contribute raw pixel likelihoods through the tag-projection model. Practical real-time estimators are then derived as structured approximations: a camera-specific inversion layer may solve a local nonlinear subproblem, but the main recursive estimator should assimilate a prior-free visual likelihood or information message rather than a prior-conditioned camera posterior reused as a new measurement. Pose pseudo-measurements are retained only as a clearly weaker fallback interface. The report explains Lie-group state representation, tangent-space Gaussians, quaternion storage versus local error coordinates, curve priors and their discrete state-space forms, pixel-level visual models, prior-removal message construction, event-driven Kalman approximations, tag initialization and maintenance, and the exact-versus-approximate treatment of first-frame setup, new-tag birth, and tag reappearance. It closes with parameter guidance, implementation architecture, probabilistic failure modes, and an appendix of prior samples that visualize the motion assumptions encoded by the model.

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How to read this report

The report is organized from modeling assumptions to practical estimators. A reader who knows basic linear algebra and probability can follow the main line without first knowing Lie theory or visual-inertial odometry.

1. First read the frame convention and pose geometry. The symbols T_{AB} , Exp , and Log are used everywhere else.
2. Then read the Bayesian target: what is unknown, what is observed, and why a camera posterior is not the same thing as a fresh measurement.
3. Next read the prior and likelihood sections. They explain which assumptions come from dynamics, which come from sensors, and which are engineering approximations.
4. Finally read the inference and implementation sections. They turn the exact posterior into practical MAP, EKF/IEKF, message-passing, or fallback pose-summary algorithms.

The notation tables are part of the exposition, not appendices. They are intended to keep each section self-contained.

Chapter 1

Foundations and Problem Setup

Roadmap. This chapter fixes frames and pose notation, states the Bayesian target, introduces the trajectory and map priors, writes the IMU and camera likelihoods, and only then derives the exact posterior and its numerical approximations. The order is deliberate: geometry first, probability model second, algorithms third.

1.1 Geometry, notation, and state representation

New notation	Meaning
$\mathbb{R}^n$	Euclidean vector space. Positions, velocities, biases, and local error coordinates live in some $\mathbb{R}^n$.
$\text{SO}(3)$	Rotation group in 3D. Elements are proper orthogonal matrices and represent orientation.
$\text{SE}(3)$	Rigid-motion group in 3D. Elements represent a full pose made of rotation and translation.
$\mathfrak{so}(3), \mathfrak{se}(3)$	Lie algebras / tangent spaces associated with $\text{SO}(3)$ and $\text{SE}(3)$.
$\mathbb{S}^3$	Unit-quaternion manifold. A quaternion $q \in \mathbb{S}^3$ is another way to represent a rotation.
Π_0	Prior law on the unknown trajectory curve.
$\hat{\cdot}, \tilde{\cdot}, \bar{\cdot}, \delta(\cdot)$	Estimated quantity, direct noisy measurement, whole curve, and local perturbation / error-state notation.

This opening section fixes the geometric language used throughout the report. Positions, velocities, and sensor biases are Euclidean variables, but orientations and rigid poses are not. The estimator therefore mixes ordinary vectors with manifold-valued states, and all later priors, residuals, and covariance statements inherit that choice.

1.1.1 Euclidean variables versus pose variables

New notation	Meaning
$p, v, b_g, b_a \in \mathbb{R}^3$	Position, velocity, gyroscope bias, and accelerometer bias.
$R \in \text{SO}(3)$	Rotation matrix.
$T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \in \text{SE}(3)$	Homogeneous rigid transform.
W, B, C	World, body/IMU, and camera frames.

A vector in $\mathbb{R}^3$ can be added, subtracted, and averaged directly. For example, if $p_1, p_2 \in \mathbb{R}^3$ are two positions, then $p_2 - p_1$ is another vector in $\mathbb{R}^3$. This is why Gaussians on positions or velocities are straightforward. A rotation is different. The set of all valid 3D rotations is

$$\text{SO}(3) = \{R \in \mathbb{R}^{3 \times 3} : R^\top R = I, \det R = 1\}. \quad (1.1)$$

This is not a linear vector space: if $R_1, R_2 \in \text{SO}(3)$, then the arithmetic average $\frac{1}{2}(R_1 + R_2)$ is usually *not* a valid rotation matrix. Similarly, a full rigid pose belongs to

$$\text{SE}(3) = \left\{ \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} : R \in \text{SO}(3), p \in \mathbb{R}^3 \right\}, \quad (1.2)$$

Notation	Meaning	Interpretation used in this note
$\bar{x}(t)$	Continuous-time unknown state curve	The mathematically primary unknown; the prior is a probability law on the whole curve, not only on one sampled state.
x_k	Discrete-time sampled state at time t_k	The state used by an online filter after discretization or at selected event times.
e_n, τ_n, s_n	Event label, event time, and event type	Sensor events are processed in chronological order; each event is either IMU or camera.
$\tilde{y}$	Direct noisy sensor reading	We use a tilde for raw sensor measurements such as IMU packets or detected pixels.
$\hat{x}$	Estimate or reduced summary	We use a hat for quantities inferred from another inverse problem, e.g. a local camera-pose estimate inside the camera inversion layer.
δx	Small local perturbation or error state	The coordinates in which Gaussian priors and Gaussian posteriors are written in EKF/IEKF-type approximations.
T_{AB}	Pose of frame B expressed in frame A	It maps coordinates from frame B to frame A ; R_{AB} and p_{AB} are its rotation and translation blocks.
T_1, T_2	Generic elements of SE(3)	Used when discussing generic pose composition or relative pose error.
$A^{-1}, A^\top$	Inverse and transpose	For poses, T^{-1} is the inverse rigid motion; for matrices, $A^\top$ is the transpose.
Exp, Log	Exponential and logarithm maps	They move between a Lie group such as SO(3) or SE(3) and local vector coordinates in its tangent space / Lie algebra.
$[\cdot]_\times$ and $(\cdot)^\wedge$	Skew / twist operators	They convert 3D rotation vectors or 6D pose twists into matrix form so that group updates can be written by matrix exponentials.
$q \in \mathbb{S}^3$	Unit quaternion	Alternative representation of orientation. In filtering one usually stores q but keeps covariance on a 3D local rotation error, not on the four quaternion coordinates.
Y_k	Raw visual data at frame k	Detected tag corners, tag ids, and per-corner quality information.
λ_k or $(\Delta\lambda_k, \Delta\eta_k)$	Prior-free visual message	The preferred modular interface from the camera inversion layer to the recursive estimator.
$z_k^v = (\hat{T}_{WC,k}, R_{v,k})$	Visual pose pseudo-measurement	A weaker fallback visual interface retained for comparison and implementation shortcuts.
$M = \{T_{WT_j}\}$	Static map of auxiliary tags	Unknown but stationary non-anchor tag poses; estimated jointly in the full visual inverse problem or by a front-end map-maintenance module.

Table 1.1: Global notation and decoration conventions. This table is meant to make the rest of the note readable without having to infer the meaning of hats, tildes, transform indices, or Lie-group operators from context.

which is again not a linear space. This is the reason that pose errors in the rest of the note are written through relative transforms and logarithms, not by subtracting the entries of two 4×4 matrices.

The transform convention used throughout the note is the robotics convention

$$T_{AB} = \begin{bmatrix} R_{AB} & p_{AB} \\ 0 & 1 \end{bmatrix}, \quad (1.3)$$

meaning “pose of frame B expressed in frame A .” If a point has coordinates x_B in frame B , then its coordinates in frame A are

$$\begin{bmatrix} x_A \\ 1 \end{bmatrix} = T_{AB} \begin{bmatrix} x_B \\ 1 \end{bmatrix}. \quad (1.4)$$

Thus $T_{AB}^{-1} = T_{BA}$. When the notation T_1, T_2 is used later, it simply means generic poses in SE(3) without specifying their frame labels.

This convention fixes the camera–IMU extrinsic notation used later. The transform T_{BC} is the pose of the camera frame C expressed in the body/IMU frame B , so it maps camera coordinates into body coordinates. Its inverse $T_{CB} = T_{BC}^{-1}$ maps body coordinates into camera coordinates. Therefore

$$T_{WC} = T_{WB}T_{BC}, \quad T_{WB} = T_{WC}T_{CB}.$$

1.1.2 Lie groups, operations, and local coordinates

New notation	Meaning
$\mathfrak{so}(3)$	Tangent space of SO(3) at the identity. It consists of 3×3 skew-symmetric matrices.
$\mathfrak{se}(3)$	Tangent space of SE(3) at the identity. It is the 6D space of twists.
$[\phi]_\times$	Skew-symmetric matrix built from $\phi \in \mathbb{R}^3$.
$\xi^\wedge$	Matrix form of a pose perturbation $\xi \in \mathbb{R}^6$.
$(\cdot)^\vee$	Inverse of the skew / twist matrix operator, returning vector coordinates.
Exp, Log	Maps between the group and local vector coordinates.
ϕ, ξ	Local rotation error and local pose error coordinates.

A group is a set G with a binary operation, written here as multiplication, satisfying four properties: closure ($g_1 g_2 \in G$ for $g_1, g_2 \in G$), associativity ($((g_1 g_2) g_3 = g_1 (g_2 g_3))$), an identity element e with $eg = ge = g$, and an inverse g^{-1} for every element. A Lie group is a group that is also a smooth manifold, with multiplication and inversion varying smoothly with the group element. In this report the concrete Lie groups are $\text{SO}(3)$ for orientation and $\text{SE}(3)$ for full rigid pose. What matters in practice is that one can compose and invert poses globally, while still doing calculus and Gaussian approximation in a local vector space. That local vector space is the tangent space at the identity, also called the Lie algebra.

The group properties are concrete here, not abstract decoration. If $R_1, R_2 \in \text{SO}(3)$, then

$$(R_1 R_2)^\top (R_1 R_2) = I, \quad \det(R_1 R_2) = \det(R_1) \det(R_2) = 1, \quad (1.5)$$

so $R_1 R_2 \in \text{SO}(3)$. The identity is I , the inverse is $R^{-1} = R^\top$, and associativity is inherited from matrix multiplication. The orthogonality and determinant constraints define a smooth three-dimensional manifold inside $\mathbb{R}^{3 \times 3}$, and matrix multiplication / transpose are smooth operations on it. Thus $\text{SO}(3)$ is a Lie group. For poses, multiplication and inversion are

$$\begin{bmatrix} R_1 & p_1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_2 & p_2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} R_1 R_2 & p_1 + R_1 p_2 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} R^\top & -R^\top p \\ 0 & 1 \end{bmatrix}. \quad (1.6)$$

Closure follows because $R_1 R_2 \in \text{SO}(3)$ and $p_1 + R_1 p_2 \in \mathbb{R}^3$; the identity is I_4 ; associativity again comes from matrix multiplication; and the inverse has the same rigid-transform form. Since $\text{SE}(3)$ is the smooth semidirect-product manifold $\text{SO}(3) \ltimes \mathbb{R}^3$ and these formulas are smooth in (R, p) , $\text{SE}(3)$ is also a Lie group.

Operation	Example	Why it appears later
Identity	$I \in \text{SO}(3), I_4 \in \text{SE}(3)$	reference element for local coordinates and perturbations
Composition	$R_1 R_2, T_1 T_2$	chaining body, camera, and tag poses
Inverse	$R^{-1} = R^\top, T^{-1}$	frame changes and relative transforms
Exponential map	$\text{Exp}([\phi]_\times), \text{Exp}(\xi^\wedge)$	injecting local updates back onto the manifold
Logarithm map	$\text{Log}(R_1^{-1} R_2), \text{Log}(T_1^{-1} T_2)$	writing residuals in local coordinates
Retraction	$T^+ = T \text{Exp}(\xi^\wedge)$	EKF, IEKF, and Gauss–Newton correction steps

The exponential and logarithm maps are the bridge between the curved group and local vector coordinates. At the matrix level, the exponential is

$$\text{expm}(A) = \sum_{n=0}^{\infty} \frac{A^n}{n!}. \quad (1.7)$$

For a Lie group, this sends a Lie-algebra element to a group element. We use the standard robotics shorthand

$$\text{Exp}([\phi]_\times) = \text{expm}([\phi]_\times) \in \text{SO}(3), \quad \text{Exp}(\xi^\wedge) = \text{expm}(\xi^\wedge) \in \text{SE}(3), \quad (1.8)$$

and

$$\text{Log}(R) = (\text{logm}(R))^\vee \in \mathbb{R}^3, \quad \text{Log}(T) = (\text{logm}(T))^\vee \in \mathbb{R}^6, \quad (1.9)$$

where logm is a locally chosen matrix logarithm. The logarithm is local rather than globally unique: for rotations the branch is unambiguous for angles strictly below π , while an angle of π has unavoidable axis/sign ambiguities. This is acceptable for filtering and Gauss–Newton updates because residuals are kept near the identity by relinearization.

For rotations, the Lie algebra is

$$\mathfrak{so}(3) = \{\Omega \in \mathbb{R}^{3 \times 3} : \Omega^\top = -\Omega\}. \quad (1.10)$$

Every vector $\phi = [\phi_1 \ \phi_2 \ \phi_3]^\top \in \mathbb{R}^3$ corresponds to a skew-symmetric matrix

$$[\phi]_\times = \begin{bmatrix} 0 & -\phi_3 & \phi_2 \\ \phi_3 & 0 & -\phi_1 \\ -\phi_2 & \phi_1 & 0 \end{bmatrix}. \quad (1.11)$$

This is why $\mathbb{R}^3$ is used as the minimal local coordinate system for small rotation errors. With $\theta = \|\phi\|$, the rotation exponential has the Rodrigues form

$$\text{Exp}([\phi]_\times) = I + \frac{\sin \theta}{\theta} [\phi]_\times + \frac{1 - \cos \theta}{\theta^2} [\phi]_\times^2, \quad (1.12)$$

where the coefficients are interpreted by their Taylor limits when $\theta \rightarrow 0$. Conversely, for $R \in \text{SO}(3)$ with rotation angle $\theta \in (0, \pi)$,

$$\text{Log}(R) = \left(\frac{\theta}{2 \sin \theta} (R - R^\top) \right)^\vee, \quad \theta = \cos^{-1} \left(\frac{\text{tr}(R) - 1}{2} \right). \quad (1.13)$$

The small-angle limit of (1.13) is simply $\text{Log}(R) \approx \text{vee}(\frac{1}{2}(R - R^\top))$. A small orientation perturbation $\delta\theta \in \mathbb{R}^3$ updates a nominal rotation by

$$R^+ = R \text{Exp}([\delta\theta]_\times). \quad (1.14)$$

For a full pose, the local perturbation is a twist

$$\xi = \begin{bmatrix} \rho \\ \phi \end{bmatrix} \in \mathbb{R}^6, \quad (1.15)$$

with matrix form

$$\xi^\wedge = \begin{bmatrix} [\phi]_\times & \rho \\ 0 & 0 \end{bmatrix} \in \mathfrak{se}(3). \quad (1.16)$$

Its exponential can be written explicitly, with the same $\theta = \|\phi\|$, as

$$\text{Exp}(\xi^\wedge) = \begin{bmatrix} \text{Exp}([\phi]_\times) & J(\phi)\rho \\ 0 & 1 \end{bmatrix}, \quad J(\phi) = I + \frac{1 - \cos \theta}{\theta^2} [\phi]_\times + \frac{\theta - \sin \theta}{\theta^3} [\phi]_\times^2, \quad (1.17)$$

again with Taylor limits at $\theta = 0$. Thus the translational block of a pose increment is not merely ρ when there is a finite rotation; it is coupled through the left Jacobian $J(\phi)$. Conversely, if

$$T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \in \text{SE}(3), \quad (1.18)$$

then the local logarithm is obtained by

$$\phi = \text{Log}(R), \quad \rho = J(\phi)^{-1}p, \quad \text{Log}(T) = \begin{bmatrix} \rho \\ \phi \end{bmatrix}. \quad (1.19)$$

Then a pose update is written as

$$T^+ = T \text{Exp}(\xi^\wedge). \quad (1.20)$$

The logarithm map does the reverse: it converts a small relative transform into local vector coordinates. For two poses $T_1, T_2 \in \text{SE}(3)$, the relative pose error is

$$\xi = \text{Log}(T_1^{-1}T_2) \in \mathbb{R}^6. \quad (1.21)$$

This is the mathematically correct replacement for “ $T_2 - T_1$ ” when the unknown is a rigid pose. Throughout the report we use this same right-multiplicative convention: nearby poses are represented as $T = \hat{T} \text{Exp}(\xi^\wedge)$, and local residuals are written with the corresponding right-invariant relative transform.

This machinery cannot be avoided in the present problem. A Gaussian density such as $\mathcal{N}(m, P)$ is defined on a vector space, not on a curved manifold. Therefore the posterior over orientation or pose cannot literally be represented as a Gaussian over the raw entries of R , T , or a quaternion while still respecting the rotation constraints. The standard construction is to keep the *nominal state* on the manifold but represent uncertainty in local Euclidean coordinates. In other words, the posterior is approximated by

$$T = \hat{T} \text{Exp}(\xi^\wedge), \quad \xi \sim \mathcal{N}(0, P_T), \quad (1.22)$$

or, for orientation alone,

$$R = \hat{R} \text{Exp}([\delta\theta]_\times), \quad \delta\theta \sim \mathcal{N}(0, P_R). \quad (1.23)$$

Lie groups are therefore not cosmetic notation. They are what keeps composition, residuals, linearization, and Gaussian uncertainty mutually consistent. The same local-coordinate language reappears in the trajectory prior, in visual residuals, in the EKF error state, and in factor-graph updates [1, 2].

1.1.3 Quaternions: storage choice, same local geometry

New notation	Meaning
$q = [q_w \ q_x \ q_y \ q_z]^\top$	Quaternion written as one scalar part and a 3D vector part.
$q \in \mathbb{S}^3$	Unit quaternion satisfying $\ q\ = 1$.
$q \sim -q$	Double-cover property: q and $-q$ represent the same physical rotation.
$\otimes$	Quaternion product.
δq	Small quaternion correction built from a 3D local rotation error.

Many implementations do not store a rotation matrix directly. They store a unit quaternion

$$q \in \mathbb{S}^3 = \{q \in \mathbb{R}^4 : \|q\| = 1\}. \quad (1.24)$$

Quaternions are a convenient storage format: they are compact, easy to normalize, and easy to propagate numerically. They do *not*, however, turn orientation into an ordinary 4D Euclidean state. A quaternion has a unit-norm constraint and the sign ambiguity $q \sim -q$, since both represent the same element of $\text{SO}(3)$.

For that reason, good filters do *not* place a naive 4D Gaussian directly on the quaternion coordinates. They follow exactly the same local-coordinate construction as with rotation matrices: the state stores a nominal quaternion $\hat{q}$, while uncertainty lives in a 3D rotation error $\delta\theta \in \mathbb{R}^3$. A small correction is converted into a unit quaternion; to first order one may write

$$\delta q \approx \begin{bmatrix} 1 \\ \frac{1}{2}\delta\theta \end{bmatrix} \quad \text{for small } \delta\theta, \quad (1.25)$$

and the nominal quaternion is updated consistently with the right-multiplicative convention above, e.g.

$$q^+ = q \otimes \delta q. \quad (1.26)$$

The quaternion representation is therefore subordinate to the Lie-group story, not a competing alternative to it. One stores orientation by a unit quaternion for convenience, but Bayesian linearization, covariance propagation, and Kalman correction still live in a 3D tangent coordinate. This is the standard multiplicative-quaternion or error-state-quaternion viewpoint [1, 3].

1.1.4 Why the unknown is really a random curve

New notation	Meaning
$\bar{x}(t)$	Continuous-time trajectory state.
Γ_T	Admissible set of such curves on $[0, T]$.
Π_0	Prior law on the curve.

The unknown in trajectory estimation is not a single pose and not a single finite vector. It is a time-indexed object: a curve that assigns a state to every time $t \in [0, T]$. The mathematically clean Bayesian object is therefore a probability measure on a path space. In this report the continuous-time state is written as

$$\bar{x}(t) = (R(t), p(t), v(t), \omega(t), b_g(t), b_a(t)), \quad (1.27)$$

where orientation is manifold-valued and the remaining variables are Euclidean. A convenient admissible path space is

$$\begin{aligned} \Gamma_T = \left\{ \bar{x} : [0, T] \rightarrow \text{SO}(3) \times (\mathbb{R}^3)^5 \mid v, \omega, b_g, b_a \in C([0, T]), \right. \\ \left. p, R \text{ are absolutely continuous,} \right. \\ \left. \dot{p} = v, \dot{R} = R[\omega]_\times \text{ a.e.} \right\}. \end{aligned} \quad (1.28)$$

The exact function-space choice is not the main point. The point is that the prior Π_0 is defined on whole curves and encodes which trajectories are plausible before any data are seen [4, 5].

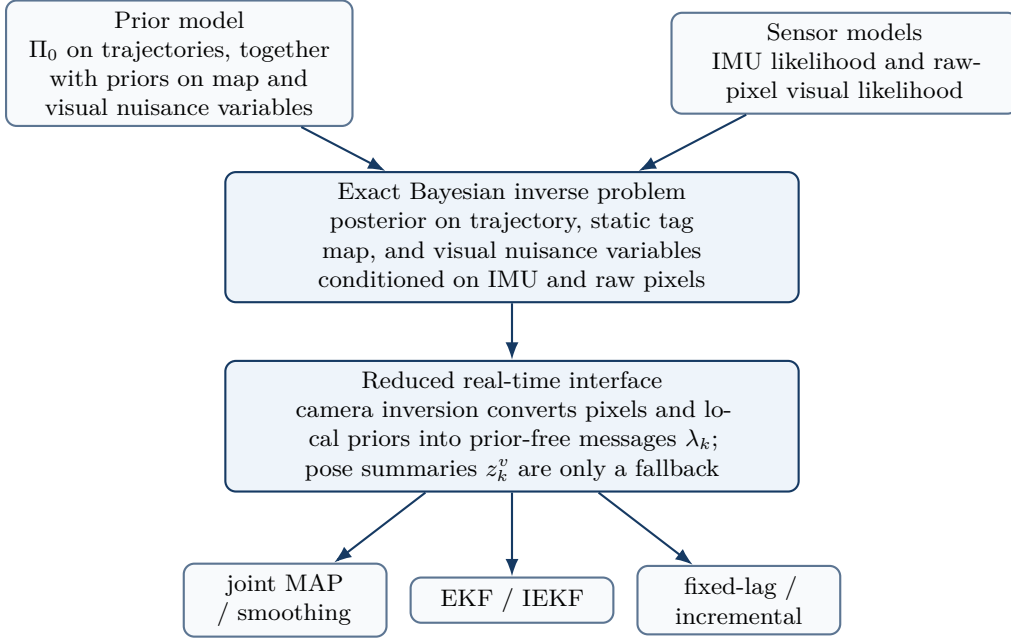


Figure 1.1: Conceptual hierarchy. The reference model is a posterior on trajectory, map, and nuisance variables; the real-time estimator works with condensed front-end summaries and then approximates that reduced problem numerically.

1.2 Problem setup and measurement interface

New notation	Meaning
W, B, C	World, body/IMU, and camera coordinate frames.
e_n	Event label in chronological order.
τ_n	Timestamp of event e_n .
$s_n \in \{\text{imu, cam}\}$	Event type: IMU packet or camera frame.
u_n	IMU packet at event n , containing measured angular rate and specific force.
Y_k	Raw image-side data at frame k , i.e. detected tag corners and quality metadata.
$\bar{p}_k^-(x_k, M)$	Predictive joint distribution of current body state and static map at camera time t_k .
λ_k	Prior-free visual likelihood / information message extracted from camera inversion.
z_k^v	Reduced fallback pose pseudo-measurement, with $z_k^v = (\hat{T}_{WC,k}, R_{v,k})$.
$\mathcal{J}_{\text{aux}}$	Index set of non-anchor tags whose world poses are unknown but stationary.
M	Static auxiliary-tag map $M = \{T_{WT_j}\}_{j \in \mathcal{J}_{\text{aux}}}$.
T_{BC}	Known pose of the camera frame expressed in the IMU/body frame; it maps camera coordinates to body coordinates.

We assume a rigid body carrying an IMU and a camera. The world frame W is anchored by at least one known calibration pattern. In the multi-tag case emphasized later, tag $j = 0$ is the anchor and its world pose T_{WT_0} is known (often chosen as the identity pose), while the remaining tags $j \in \mathcal{J}_{\text{aux}}$ are only known to be stationary. The IMU/body frame is B , the camera frame is C , and the camera extrinsic is

$$T_{BC} = \begin{bmatrix} R_{BC} & t_{BC} \\ 0 & 1 \end{bmatrix} \in \text{SE}(3) \quad (1.29)$$

which is assumed known from an earlier calibration step. By the convention above, T_{BC} maps camera-frame coordinates into body-frame coordinates and $T_{CB} = T_{BC}^{-1}$ maps body-frame coordinates into camera-frame coordinates.

The sensor stream is asynchronous. Measurements are assimilated one event at a time, not in artificially synchronized camera–IMU pairs. Let

$$0 \leq \tau_1 < \tau_2 < \dots < \tau_N, \quad s_n \in \{\text{imu, cam}\}, \quad (1.30)$$

and let $\mathcal{D}_{1:n}$ denote all sensor events observed up to event e_n . If $s_n = \text{imu}$, the event carries an IMU packet

$$u_n = (\tilde{\omega}_n, \tilde{f}_n, \tau_n), \quad (1.31)$$

where $\tilde{\omega}_n$ is measured angular velocity and $\tilde{f}_n$ is measured specific force. If $s_n = \text{cam}$, the event is a camera frame acquired at time $t_k = \tau_n$ with raw visual data

$$Y_k = \{\tilde{u}_{i,k}^{(j)}, s_{i,k}^{(j)}, \text{tag id } j\}_{j \in \mathcal{V}_k, i=1, \dots, m_j}, \quad (1.32)$$

where $\tilde{u}_{i,k}^{(j)} \in \mathbb{R}^2$ is the detected pixel coordinate of corner i of tag j , $s_{i,k}^{(j)}$ denotes detector quality metadata such as margin or score, and $\mathcal{V}_k$ is the set of visible tags at frame k . In the simplest anchor-only case the front-end computes

$$x(t) = (R_{WB}(t), p_{WB}(t), v_W(t), b_g(t), b_a(t)),$$

and the static map is

$$M = \{T_{WT_j}\}_{j \in \mathcal{J}_{\text{aux}}}.$$

The anchor pose T_{WT_0} is fixed and known. The exact Bayesian target is therefore a joint posterior over the dynamic body trajectory and the static auxiliary-tag map conditioned on the full event stream.

The central modeling distinction in this report is between three objects:

1. the predictive prior at a camera time, $p_k^-(x_k, M) = p(x(t_k), M \mid \mathcal{D}_{<k})$,
2. the raw pixel likelihood $p(Y_k \mid x_k, M)$,
3. the posterior after the camera event, $p_k^+(x_k, M) \propto p(Y_k \mid x_k, M) p_k^-(x_k, M)$.

This distinction matters because a modular camera subproblem may use the predictive prior internally. If it returns a posterior and the main estimator then treats that posterior as a new independent measurement, the prior has been counted twice. The defensible modular interface is therefore a *prior-free visual message* λ_k , or equivalently an information increment $(\Delta\Lambda_k, \Delta\eta_k)$ extracted from the camera inversion. A reduced pose pseudo-measurement

$$z_k^v = (\hat{T}_{WC,k}, R_{v,k}) \quad (1.33)$$

is retained later as a weaker fallback approximation, useful only when one accepts that pose-map correlation and prior reuse are being simplified away.

This event-driven viewpoint keeps the modeling honest while still allowing modular implementations. The camera inversion layer is free to use nonlinear optimization, robust losses, sliding windows, or visible-tag subsets internally. The main recursive estimator must distinguish a likelihood contribution from a posterior summary whenever the camera block used the predicted prior internally. The literature comparison is collected near the end of the report, after the model and approximation hierarchy have been defined.

1.3 Bayesian model for the trajectory prior

New notation	Meaning
$\bar{x}(t)$	Continuous-time latent state. In the prior model we use $\bar{x}(t) = (R(t), p(t), v(t), \omega(t), b_g(t), b_a(t))$.
w_v, w_ω	White-noise drives for translational and rotational smoothness.
w_{bg}, w_{ba}	White-noise drives for gyro and accelerometer bias drift.
$Q_v, Q_\omega, Q_{bg}, Q_{ba}$	Spectral-density matrices controlling roughness of the prior.
m_0, P_0	Mean and covariance of the initial-state prior.
Σ_{map}	Optional prior covariance for a sequence-level visual bias.

A prior on the curve $\bar{x}(\cdot)$ must say which trajectories are plausible before seeing the data. In this report the prior is introduced as the law of a stochastic process generated by a stochastic differential equation. This section is the first half of the Bayesian model: it explains the trajectory law before any sensor likelihood is applied. Write

$$\bar{x}(t) = (R(t), p(t), v(t), \omega(t), b_g(t), b_a(t)), \quad (1.34)$$

with kinematic relations

$$\dot{p}(t) = v(t), \quad \dot{R}(t) = R(t) [\omega(t)]_\times. \quad (1.35)$$

A smoothness prior is then generated by

$$\dot{v}(t) = w_v(t), \quad w_v(t) \sim \mathcal{N}(0, Q_v \delta(t - t')), \quad (1.36)$$

$$\dot{\omega}(t) = w_\omega(t), \quad w_\omega(t) \sim \mathcal{N}(0, Q_\omega \delta(t - t')), \quad (1.37)$$

$$\dot{b}_g(t) = w_{bg}(t), \quad w_{bg}(t) \sim \mathcal{N}(0, Q_{bg} \delta(t - t')), \quad (1.38)$$

$$\dot{b}_a(t) = w_{ba}(t), \quad w_{ba}(t) \sim \mathcal{N}(0, Q_{ba} \delta(t - t')). \quad (1.39)$$

Under the continuous-time white-noise idealization, these equations are interpreted in integral form and define a probability law $\Pi_0 = \text{Law}(\bar{x})$ on Γ_T . Translational acceleration, angular acceleration, and bias drift are modeled through zero-mean stochastic drives. The matrices $Q_v, Q_\omega, Q_{bg}, Q_{ba}$ are therefore not measurement-noise covariances; they are *hyperparameters of the prior law on trajectories*. They control how rough the trajectory is allowed to be. Small Q_v means the prior expects nearly constant velocity. Large Q_v allows aggressive translational maneuvers. Small Q_{bg} and Q_{ba} imply nearly constant biases, while larger values allow faster drift.

Because these are continuous-time spectral densities, their discrete effect depends on the integration interval. For example, a velocity-drive covariance over a short interval is proportional to $Q_v \Delta t$, and a bias random-walk increment covariance is proportional to $Q_{bg} \Delta t$ or $Q_{ba} \Delta t$. This scaling is the reason that continuous-time noise densities should not be copied directly into a per-frame covariance without accounting for the sample interval.

A commonly used formal quadratic action associated with this prior is

$$\mathcal{J}_{\text{prior}}^{\text{formal}}(\bar{x}) = \frac{1}{2} \int_0^T \left(\|\dot{v}\|_{Q_v^{-1}}^2 + \|\dot{\omega}\|_{Q_\omega^{-1}}^2 + \|\dot{b}_g\|_{Q_{bg}^{-1}}^2 + \|\dot{b}_a\|_{Q_{ba}^{-1}}^2 \right) dt, \quad (1.40)$$

subject to the kinematic constraints on R and p . Equation (1.40) should be read as a formal shorthand or MAP/discretized surrogate on admissible paths; it is *not* a literal density of Π_0 with respect to a path-space Lebesgue measure.

1.3.1 Why these prior components are reasonable

New notation	Meaning
Q_v	Controls translational roughness. Large values allow larger acceleration-like changes in velocity.
Q_ω	Controls rotational roughness. Large values allow larger angular-acceleration-like changes in angular velocity.
Q_{bg}, Q_{ba}	Control how quickly gyro and accelerometer biases are allowed to drift.
ρ_p, ρ_R, Q_c	Parameters of colored visual-noise state, introduced later.

The rationale for the prior components is physical. Over short video intervals, position and orientation do not jump discontinuously, so a constant-velocity / constant-angular-velocity prior is a sensible default. The white-noise drives in (1.39) provide a simple smoothness model that still admits real maneuvers. Bias random walks are used because MEMS bias drift is slow, cumulative, and not known a priori; a random walk is a common tractable model for a slowly varying nuisance offset and integrates well with filters and smoothers. The same bias-random-walk nuisance model is used in Kalibr and in standard visual-inertial formulations such as MSCKF-style filters and preintegrated optimization methods [6–8].

The prior may also include nuisance terms not tied directly to rigid-body dynamics. A sequence-level visual bias can be represented as

$$b_{\text{map}} \sim \mathcal{N}(0, \Sigma_{\text{map}}), \quad (1.41)$$

where $b_{\text{map}} \in \mathbb{R}^6$ captures slowly varying or constant pose offsets due to imperfect anchor geometry, imperfect camera calibration, or tag-model mismatch. If the anchor and camera model are treated as exact, set $b_{\text{map}} = 0$.

1.3.2 From the curve prior to the state-space prior used by a Kalman filter

New notation	Meaning
$\tilde{x}_k$	Sampled extended state at frame time t_k , here $\tilde{x}_k = (R_k, p_k, v_k, \omega_k, b_{g,k}, b_{a,k})$.
x_k	Reduced practical filter state, usually $x_k = (R_k, p_k, v_k, b_{g,k}, b_{a,k})$ after conditioning on IMU packets.
$p(\tilde{x}_{k+1} \tilde{x}_k)$	One-step Markov transition induced by the discretized SDE.
θ_{prior}	Collection of prior hyperparameters.

A Kalman filter does not manipulate the full path measure directly. A sampled SDE is Markov on an extended state that contains all variables needed to close the dynamics. For the prior above that extended state is

$$\tilde{x}_k = (R_k, p_k, v_k, \omega_k, b_{g,k}, b_{a,k}), \quad (1.42)$$

and its sampled law factorizes as

$$p(\tilde{x}_{0:N}) = p(\tilde{x}_0) \prod_{k=0}^{N-1} p(\tilde{x}_{k+1} | \tilde{x}_k; \theta_{\text{prior}}), \quad (1.43)$$

and hyperparameter set

$$\theta_{\text{prior}} = \{m_0, P_0, Q_v, Q_\omega, Q_{bg}, Q_{ba}, \Sigma_{\text{map}}, \rho_p, \rho_R, Q_c\}. \quad (1.44)$$

In many practical VIO filters the angular velocity is not carried as a separate state because the gyroscope packets are treated as observed inputs in the propagation step. After conditioning on the IMU packets between frames, one obtains the reduced transition kernel $p(x_{k+1} | x_k, \mathcal{U}_k)$ for

$$x_k = (R_k, p_k, v_k, b_{g,k}, b_{a,k}). \quad (1.45)$$

Thus the Markov prior used by a filter is not a different physical prior; it is a sampled and, if desired, measurement-conditioned form of the same prior-on-curves.

A convenient initial prior is specified locally around a nominal initial state m_0 : rotational components are perturbed multiplicatively, Euclidean components additively, and the local error covariance is P_0 . The nominal state m_0 is usually derived from the first reliable visual pose, a short stationary IMU segment, or trusted setup geometry, and P_0 expresses how uncertain that initialization is. A common block structure is

$$P_0 = \text{diag}(\sigma_{R0}^2 I_3, \sigma_{p0}^2 I_3, \sigma_{v0}^2 I_3, \sigma_{bg0}^2 I_3, \sigma_{ba0}^2 I_3). \quad (1.46)$$

1.3.3 Prior for stationary but initially unknown auxiliary tags

New notation	Meaning
$M = \{T_{WT_j}\}_{j \in \mathcal{J}_{\text{aux}}}$	Static map of auxiliary tag poses.
$\pi_M(M)$	Prior law on the auxiliary-tag map.
$\mathcal{W}$	Coarse workspace region in which auxiliary tags are believed to lie.
$\mu_{M,j}, \Sigma_{M,j}$	Mean and covariance of a local Gaussian prior for auxiliary tag j after anchored initialization.

When only one tag is anchored, the prior is not only a law on the body trajectory. It also includes a prior on the unknown but stationary auxiliary-tag map

$$M = \{T_{WT_j}\}_{j \in \mathcal{J}_{\text{aux}}}. \quad (1.47)$$

Before any anchored observation of auxiliary tag j , the mathematically honest prior is broad: we only know that the tag is stationary and that it lies somewhere in the lab workspace. A minimal prior is therefore a support prior of the form

$$\pi_M(dM) \propto \prod_{j \in \mathcal{J}_{\text{aux}}} \mathbf{1}\{p_{WT_j} \in \mathcal{W}\} \pi_{R,j}(dR_{WT_j}), \quad (1.48)$$

where $\mathcal{W} \subset \mathbb{R}^3$ is a coarse workspace box and $\pi_{R,j}$ is an orientation prior reflecting any known mounting convention, for example “tags are roughly vertical on walls” or “tags are roughly horizontal on cube tops.”

Once an auxiliary tag is first seen simultaneously with the anchor, one obtains an anchored initialization $\mu_{M,j} = \hat{T}_{WT_j}^{\text{init}}$. After that point the prior can be tightened and written in local Lie coordinates as

$$T_{WT_j} = \mu_{M,j} \text{Exp}(\eta_{M,j}^\wedge), \quad \eta_{M,j} \sim \mathcal{N}(0, \Sigma_{M,j}), \quad (1.49)$$

with small or slowly shrinking covariance $\Sigma_{M,j}$. Again, the Gaussian lives in the local coordinate $\eta_{M,j} \in \mathbb{R}^6$, not directly on $\text{SE}(3)$. In a batch or fixed-lag front-end this is the natural way to represent “stationary but initially unknown.” In an online filter, one either augments the state with these nearly-static variables or lets a separate front-end map-maintenance module update $\widehat{M}_k$ and its uncertainty.

The key conceptual point is that the anchor removes the global $\text{SE}(3)$ gauge freedom, while the stationary auxiliary-tag prior regularizes the remaining map estimation problem. Without the anchor, the camera trajectory and the whole map could drift together by the same rigid transform without changing any pixel prediction.

1.4 Bayesian model for the IMU channel

New notation	Meaning
$\tilde{\omega}(t)$	Measured angular velocity.
$\tilde{a}(t)$	Measured specific force.
g_W	Gravity in the world frame.
n_g, n_a	Zero-mean gyro and accelerometer white noise.
b_g, b_a	Slowly varying gyro and accelerometer biases.
R_g, R_a	Continuous-time gyro and accelerometer noise intensities used in the formal likelihood notation; after discretization they contribute to Q_k .

The IMU returns two signals: angular rate from the gyroscope and specific force from the accelerometer. The gyroscope model is

$$\tilde{\omega}(t) = \omega(t) + b_g(t) + n_g(t), \quad (1.50)$$

where $\omega(t)$ is the true angular velocity expressed in the body frame, $b_g(t)$ is a slowly varying bias, and $n_g(t)$ is fast zero-mean noise.

The accelerometer is more subtle. It does not measure world-frame acceleration directly. It measures the non-gravitational specific force expressed in the sensor frame:

$$\tilde{a}(t) = R_{BW}(t)(\ddot{p}_W(t) - g_W) + b_a(t) + n_a(t). \quad (1.51)$$

Here $\ddot{p}_W(t)$ is translational acceleration in the world frame and g_W is the gravity vector in that same frame. The term $R_{BW}(t)(\ddot{p}_W(t) - g_W)$ converts the world-frame specific force into body coordinates. This is why inertial propagation later re-adds gravity in the world frame [8, 9].

1.4.1 IMU as a likelihood on the unknown curve

New notation	Meaning
$\Phi_u(\bar{x}; y^u)$	IMU negative log-likelihood or data-misfit functional.
$\ r\ _{M^{-1}}^2$	Weighted squared norm $r^\top M^{-1} r$.

If one stays in the path-space Bayesian language, the IMU contributes a data-misfit functional on the whole curve. Under the common Gaussian white-noise idealization one writes formally

$$\Phi_u(\bar{x}; y^u) = \frac{1}{2} \int_0^T \left[\|\tilde{\omega}(t) - \omega(t) - b_g(t)\|_{R_g^{-1}}^2 + \|\tilde{a}(t) - R_{BW}(t)(\ddot{p}(t) - g_W) - b_a(t)\|_{R_a^{-1}}^2 \right] dt. \quad (1.52)$$

This is the continuous-time analogue of the familiar sum of weighted squared innovations: for any candidate trajectory curve, it quantifies how well that curve explains the measured angular rates and specific forces. As in the prior section, the white-noise notation is formal; rigorous treatments interpret it through discretization or Gaussian-measure constructions rather than through an ordinary Lebesgue density on signal space.

1.4.2 IMU as the process model used by a practical online filter

New notation	Meaning
$\mathcal{U}_k$	All IMU packets in a chosen propagation interval, often between two camera exposure times t_k and t_{k+1} .
Q_k	Effective discrete process covariance over one propagation interval.

In an online filter it is more convenient to condition on the IMU packets and use the resulting state-transition model. The corresponding continuous-time kinematics are

$$\dot{R}_{WB} = R_{WB} [\tilde{\omega} - b_g - n_g]_{\times}, \quad (1.53)$$

$$\dot{v}_W = g_W + R_{WB}(\tilde{a} - b_a - n_a), \quad (1.54)$$

$$\dot{p}_{WB} = v_W, \quad (1.55)$$

$$\dot{b}_g = w_{bg}, \quad (1.56)$$

$$\dot{b}_a = w_{ba}. \quad (1.57)$$

A first-order discrete propagation over one short interval of length Δt is

$$R_{k+1} = R_k \text{Exp}([\tilde{\omega}_k - b_{g,k}]_{\times} \Delta t), \quad (1.58)$$

$$v_{k+1} = v_k + (g_W + R_k(\tilde{a}_k - b_{a,k}))\Delta t + n_k^v, \quad (1.59)$$

$$p_{k+1} = p_k + v_k \Delta t + \frac{1}{2}(g_W + R_k(\tilde{a}_k - b_{a,k}))\Delta t^2 + n_k^p, \quad (1.60)$$

$$b_{g,k+1} = b_{g,k} + n_k^{bg}, \quad b_{a,k+1} = b_{a,k} + n_k^{ba}. \quad (1.61)$$

The discrete noise terms above are not independent free parameters; they arise from integrating n_g, n_a, w_{bg}, w_{ba} over the actual propagation interval, and their joint covariance is summarized by Q_k . In an event-driven implementation the interval may be one IMU sample or the time from the previous event to the current event. If one writes a camera-to-camera transition, it is the accumulated result of all IMU packets between the two exposure times. Equation (1.61) is therefore a first-order discretization of the IMU-conditioned state transition used in EKF-, IEKF-, or fixed-lag estimators [2, 7, 8].

1.5 Bayesian model for the visual channel

New notation	Meaning
Y_k	Raw visual data at frame k : detected corners, tag ids, and quality metadata.
q_k^-, q_k^+	Local predictive prior and local posterior used inside a camera inversion subproblem.
λ_k	Prior-free visual likelihood or information message extracted from the camera inversion.
$\Delta\Lambda_k, \Delta\eta_k$	Information-matrix and information-vector increments representing the visual contribution alone.
$z_k^v = (\hat{T}_{WC,k}, R_{v,k})$	Reduced fallback pose pseudo-measurement retained for comparison.
$P_i^{(j)}$	Corner i of tag j in tag-local coordinates.
$\tilde{u}_{i,k}^{(j)}$	Measured image location of a tag corner.
$\Sigma_{u,i,k}$	Pixel-space covariance of one detected corner.

This section makes the camera-side conditioning structure explicit. At a camera timestamp t_k there are three distinct probabilistic objects:

1. the predictive prior $p_k^-(x_k, M) = p(x(t_k), M \mid \mathcal{D}_{<k})$ supplied by the event-driven estimator,
2. the pixel likelihood $p(Y_k \mid x_k, M)$ supplied by the camera and tag projection model,
3. the posterior $p_k^+(x_k, M) \propto p(Y_k \mid x_k, M) p_k^-(x_k, M)$ after the frame is assimilated.

The exact model uses the pixel likelihood directly. A modular camera inversion layer may instead compute a local posterior and then extract a prior-free message λ_k . A pose pseudo-measurement z_k^v is only a weaker fallback, appropriate when one is willing to discard camera-map correlation and accept a cruder visual interface.

1.5.1 Pixel-level camera model

New notation	Meaning
K	Camera intrinsics matrix.
$\pi(K, \cdot)$	Camera projection function from camera-frame 3D coordinates to pixel coordinates.
$T_{WC,k}$	True camera pose in the world frame.
T_{WT_j}	World pose of tag j .
$u_{i,k}^{(j)}$	Ideal pixel coordinate predicted by the camera model.

For a visible tag j and one of its corners $P_i^{(j)}$ in tag-local coordinates, the corresponding world point is

$$P_{W,i}^{(j)} = T_{WT_j} P_i^{(j)}. \quad (1.62)$$

Given the true camera pose $T_{WC,k}$, the ideal pixel prediction is

$$u_{i,k}^{(j)} = \pi\left(K, T_{WC,k}^{-1} P_{W,i}^{(j)}\right) = \pi\left(K, T_{WC,k}^{-1} T_{WT_j} P_i^{(j)}\right). \quad (1.63)$$

The detector returns a noisy corner location

$$\tilde{u}_{i,k}^{(j)} = u_{i,k}^{(j)} + r_{i,k}^{(j)} + d_{i,k}^{(j)} + o_{i,k}^{(j)}, \quad (1.64)$$

where $r_{i,k}^{(j)}$ is the rasterization / quantization floor, $d_{i,k}^{(j)}$ is ordinary corner-localization error from blur and image noise, and $o_{i,k}^{(j)}$ is an occasional larger outlier term due to partial occlusion, decoding failure, or incorrect matching. In a Gaussian approximation one summarizes the inlier part by a covariance $\Sigma_{u,i,k}$ and handles the outlier part by gating or robust losses.

1.5.2 Front-end inverse problem when the anchor is known

New notation	Meaning
$\mathcal{V}_k$	Set of visible tags in frame k .
m_j	Number of corners used from tag j .
$\rho(\cdot)$	Optional robust loss.
W_k	Weight matrix assembled from corner covariances and robust weights.
$\xi_{C,k} \in \mathbb{R}^6$	Local camera-pose perturbation used inside the camera inversion.

If the world poses of all visible tags are already known, the per-frame camera inversion is a local pose-estimation problem. The direct pixel-side objective is still

$$\hat{T}_{WC,k} = \underset{T \in \text{SE}(3)}{\text{argmin}} \frac{1}{2} \sum_{j \in \mathcal{V}_k} \sum_{i=1}^{m_j} \rho\left(\left\|\tilde{u}_{i,k}^{(j)} - \pi(K, T^{-1} T_{WT_j} P_i^{(j)})\right\|_{\Sigma_{u,i,k}^{-1}}^2\right). \quad (1.65)$$

What changes in the present report is the interface to the main estimator. Let the camera inversion use a local prior

$$q_k^-(\xi_{C,k}) \approx \mathcal{N}(\mu_k^-, P_k^-)$$

derived from the predicted body state at the exposure time. After processing the pixels, it returns a local posterior

$$q_k^+(\xi_{C,k}) \propto p(Y_k | \xi_{C,k}) q_k^-(\xi_{C,k}) \approx \mathcal{N}(\mu_k^+, P_k^+).$$

The correct modular object is then not q_k^+ treated as a new measurement. Instead, one removes the local prior in information form:

$$\Delta\Lambda_k = (P_k^+)^{-1} - (P_k^-)^{-1}, \quad \Delta\eta_k = (P_k^+)^{-1} \mu_k^+ - (P_k^-)^{-1} \mu_k^-. \quad (1.66)$$

The resulting message

$$\lambda_k(\xi_{C,k}) \propto \exp\left(-\frac{1}{2} \xi_{C,k}^\top \Delta\Lambda_k \xi_{C,k} + \Delta\eta_k^\top \xi_{C,k}\right)$$

approximates the pixel likelihood contribution alone. Only if the camera inversion is solved with a flat prior, or if one intentionally accepts a weaker reduction, does it collapse to a pose pseudo-measurement summary

$$R_{v,k} \approx (J_k^\top W_k J_k)^{-1}, \quad z_k^v = (\hat{T}_{WC,k}, R_{v,k}), \quad (1.67)$$

which is then interpreted only as a fallback interface rather than the report's mainline model.

1.5.3 Multiple tags when only one tag is anchored

New notation	Meaning
T_{WT_0}	Known world pose of the anchor tag, often chosen as the identity pose.
$\mathcal{J}_{\text{aux}}$	Set of non-anchor tags.
$M = \{T_{WT_j}\}_{j \in \mathcal{J}_{\text{aux}}}$	Unknown but stationary auxiliary-tag map.
z_k	Local variable block containing current camera and visible auxiliary-tag perturbations.
$\xi_{T_j} \in \mathbb{R}^6$	Local perturbation of visible auxiliary tag j .
$\psi_M(M)$	Static-map regularizer or prior term.

The practically important case is that only the anchor tag $j = 0$ has known world pose. All other tags are only known to be stationary. Then the camera event is no longer a pure pose-estimation problem. It is a local joint *pose-plus-visible-map* problem. A compact local variable block is

$$z_k = (\xi_{C,k}, \{\xi_{T_j}\}_{j \in \mathcal{V}_k \cap \mathcal{J}_{\text{aux}}}),$$

where every pose perturbation lives in local Lie coordinates. The exact camera posterior at time t_k is

$$p_k^+(x_k, M) \propto p(Y_k | x_k, M) p_k^-(x_k, M), \quad (1.68)$$

and a modular camera inversion computes the corresponding local approximation

$$q_k^+(z_k) \propto p(Y_k | z_k) q_k^-(z_k).$$

If both are approximated locally by Gaussians,

$$q_k^-(z_k) = \mathcal{N}(\mu_k^-, P_k^-), \quad q_k^+(z_k) = \mathcal{N}(\mu_k^+, P_k^+),$$

then the message extracted from the frame is again the prior-removed information increment

$$\Delta\Lambda_k = (P_k^+)^{-1} - (P_k^-)^{-1}, \quad \Delta\eta_k = (P_k^+)^{-1}\mu_k^+ - (P_k^-)^{-1}\mu_k^-, \quad (1.69)$$

now defined on the coupled camera-and-visible-tag block. This is the main reason the message interface is preferable to a bare pose summary: the same pixels constrain current pose and visible tag states together, and the estimator should preserve that local coupling whenever it can.

The weaker alternative is to collapse the camera block to a pose-only summary z_k^v . That reduction can still be useful for diagnostics or for a first prototype, but it no longer represents the full visual information carried by jointly visible auxiliary tags, and it becomes unsafe if the summary was built from the same prior already used by the body filter.

1.5.4 Why the visual pose covariance changes from frame to frame

New notation	Meaning
z_k	Approximate range from camera to tag plane.
θ_k	Viewing angle between tag normal and camera optical axis.
$A_{\text{tag},k}$	Projected tag area in pixels.
β_k	Blur or image-quality penalty.
$\sigma_{\text{px},k}$	Effective corner-localization standard deviation in pixels.

The covariance $R_{v,k}$ is heteroscedastic: it changes with geometry and image quality. The reason is that the underlying PnP-type inverse problem becomes more ill-conditioned when the tag is far away, strongly tilted, small in the image, motion-blurred, or near the image border. A useful engineering model is

$$\sigma_{\text{px},k}^2 = \sigma_0^2 + \frac{\alpha_A}{A_{\text{tag},k}} + \alpha_\theta \tan^2 \theta_k + \alpha_b \beta_k^2, \quad (1.70)$$

which says that corner localization gets worse when the tag covers fewer pixels, becomes more oblique, or is more blurred. Local PnP geometry then implies approximate scaling laws such as

$$\sigma_{xy,k} \propto \frac{z_k}{f} \sigma_{\text{px},k}, \quad \sigma_{z,k} \propto \frac{z_k^2}{f \ell_{\text{eff},k}} \sigma_{\text{px},k}, \quad \sigma_{\text{rot},k} \propto \frac{z_k}{f \ell_{\text{eff},k}} \sigma_{\text{px},k}, \quad (1.71)$$

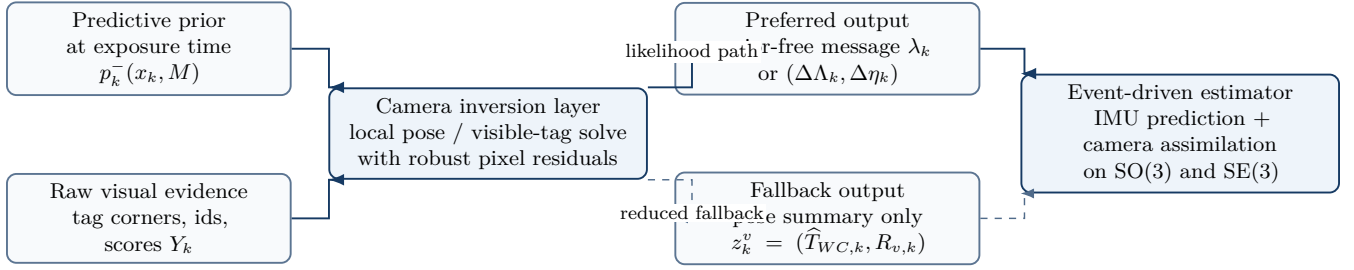


Figure 1.2: Reduced visual interface with the recommended message layer. Raw pixels and the predictive local prior enter a camera-specific inversion block. The main estimator should consume the prior-free visual message; a pose summary is retained only as a weaker fallback approximation.

where f is focal length in pixels and $\ell_{\text{eff},k}$ is the effective informative tag span in the image. Experimental AprilTag and planar-fiducial studies report that distance and obliquity degrade pose accuracy [10–12].

In the event-driven message formulation, these same geometry effects determine the local Hessian of the camera inversion and therefore the strength of the information increment $(\Delta\Lambda_k, \Delta\eta_k)$. If one falls back to a pose pseudo-measurement, they determine $R_{v,k}$. In both cases the visual contribution is heteroscedastic and must not be modeled as one fixed covariance for all frames.

1.5.5 Temporal correlation in reduced visual summaries

New notation	Meaning
c_k	Temporally correlated component of visual pose error.
b_{map}	Sequence-level static visual bias term.
Φ, Q_c	State-transition matrix and process covariance of the colored visual-noise state.

Adjacent reduced visual summaries are often correlated because the same tags, the same camera calibration, the same optical conditions, and the same front-end initialization are reused across many consecutive frames. A useful model for the weaker pose-summary interface is therefore

$$\text{Log}(\hat{T}_{WC,k}^{-1} T_{WC,k}) = b_{\text{map}} + c_k + \nu_k, \quad \nu_k \sim \mathcal{N}(0, R_{v,k}), \quad (1.72)$$

with colored term

$$c_{k+1} = \Phi c_k + w_k^c, \quad \Phi = \text{diag}(\rho_p I_3, \rho_R I_3), \quad w_k^c \sim \mathcal{N}(0, Q_c). \quad (1.73)$$

The constant term b_{map} captures slowly varying or sequence-level bias, while c_k captures short-memory temporal correlation. If one ignores these terms and pretends that the reduced visual summaries are i.i.d. Gaussian, the resulting filter is usually overconfident. Even the message interface can inherit mild correlation if successive camera subproblems share a sliding window or stale map linearization, so any reduced interface should be described as an approximation rather than as exact Bayes.

1.6 Inference structure: exact posterior and practical approximations

New notation	Meaning
$\Pi(d\bar{x}, dM, dc, db_{\text{map}} \mid \mathcal{D})$	Full posterior measure on trajectory, map, and visual nuisance variables conditioned on the event stream.
Π_{ref}	Product reference prior measure $\Pi_0 \otimes \pi_M \otimes \Pi_c \otimes \pi_{\text{map}}$.
ψ_n	Event factor: inertial transition likelihood or camera likelihood contribution.
$\tilde{\Pi}(d\bar{x}, dM, dc, db_{\text{map}} \mid y^u, \lambda)$	Reduced approximate posterior after replacing each camera frame by a prior-free message.
$\Phi(\cdot)$	Negative log likelihood or data-misfit functional.
$J(\cdot)$	Discretized negative log posterior / MAP objective.
A_k	Selector that injects a local camera message into the relevant joint-state coordinates.

At the fully Bayesian level there are four unknown objects: the continuous trajectory curve $\bar{x}(\cdot)$, the stationary auxiliary-tag map M , the temporally correlated visual-error process c , and the sequence-level visual bias

b_{map} . The clean posterior on these unknowns is a probability measure. If y^u denotes the raw IMU stream and Y the raw visual stream, then Bayes' formula is written symbolically as

$$\Pi(d\bar{x}, dM, dc, db_{\text{map}} \mid y^u, Y) \propto p(y^u \mid \bar{x}) p(Y \mid \bar{x}, M, c, b_{\text{map}}) \Pi_0(d\bar{x}) \pi_M(dM) \Pi_c(dc) \pi_{\text{map}}(db_{\text{map}}). \quad (1.74)$$

This is the most faithful statement of the inverse problem. It says: start from a prior law on trajectories and nuisance variables, then condition that law on the observed data. To make the Radon–Nikodym formula precise, introduce the product reference prior measure

$$\Pi_{\text{ref}} = \Pi_0 \otimes \pi_M \otimes \Pi_c \otimes \pi_{\text{map}}.$$

Then one may write

$$\frac{d\Pi}{d\Pi_{\text{ref}}}(\bar{x}, M, c, b_{\text{map}}) \propto \exp(-\Phi_u(\bar{x}; y^u) - \Phi_v(\bar{x}, M, c, b_{\text{map}}; Y)). \quad (1.75)$$

This notation emphasizes that the posterior is still a measure on functions, not yet a finite-dimensional Gaussian vector [4, 5].

After time discretization, the same model can be written as an event-factorized posterior. If x_n denotes the dynamic state at event time τ_n , then

$$p(x_{1:N}, M, c, b_{\text{map}} \mid \mathcal{D}_{1:N}) \propto p(x_1) p(M) p(c_1, b_{\text{map}}) \prod_{n=2}^N \psi_n, \quad (1.76)$$

with event factor

$$\psi_n = \begin{cases} p(u_n \mid x_n, x_{n-1}), & s_n = \text{imu}, \\ p(Y_k \mid x_n, M, c_n, b_{\text{map}}), & s_n = \text{cam}. \end{cases}$$

Equation (1.76) is the discrete counterpart of the path-space posterior: each physical data source contributes exactly one factor.

In the preferred reduced real-time formulation the raw pixels are *not* replaced by a camera posterior. They are replaced, frame by frame, by prior-free camera messages λ_k . The reduced approximate posterior is therefore

$$\tilde{\Pi}(d\bar{x}, dM, dc, db_{\text{map}} \mid y^u, \lambda) \propto p(y^u \mid \bar{x}) \prod_k \lambda_k(\bar{x}, M, c, b_{\text{map}}) \Pi_0(d\bar{x}) \pi_M(dM) \Pi_c(dc) \pi_{\text{map}}(db_{\text{map}}). \quad (1.77)$$

Here each λ_k is meant to approximate the visual likelihood contribution of frame k alone. The weakest reduction replaces λ_k by a pose pseudo-measurement kernel on z_k^v , but that step is an additional approximation rather than the report's mainline model.

1.6.1 Three numerical routes from the posterior to an estimator

New notation	Meaning
MCMC / SMC	Sampling-based approximation of the posterior law.
MAP	Maximum a posteriori estimate obtained by minimizing the negative log posterior.
Fixed lag	Sliding-window approximation that repeatedly solves a recent MAP problem.
EKF / IEKF	Recursive Gaussian approximation to the filtering posterior.

There are three standard ways to turn the Bayesian inverse problem into a numerical algorithm.

Route A: sample the posterior. One can approximate the full posterior in (1.74) by Markov-chain Monte Carlo or by sequential Monte Carlo. This is closest to the mathematically exact Bayesian formulation and best reflects multimodality or non-Gaussian uncertainty, but it is usually too expensive for 60 Hz online pose tracking.

Route B: compute a MAP estimate. After choosing a finite-dimensional discretization and local coordinates, one obtains a cost functional of the form

$$J(X, M, c, b_{\text{map}}) = \Phi_u(X; U) + \Phi_v(X, M, c, b_{\text{map}}; Y) + J_{\text{prior}}(X, M, c, b_{\text{map}}), \quad (1.78)$$

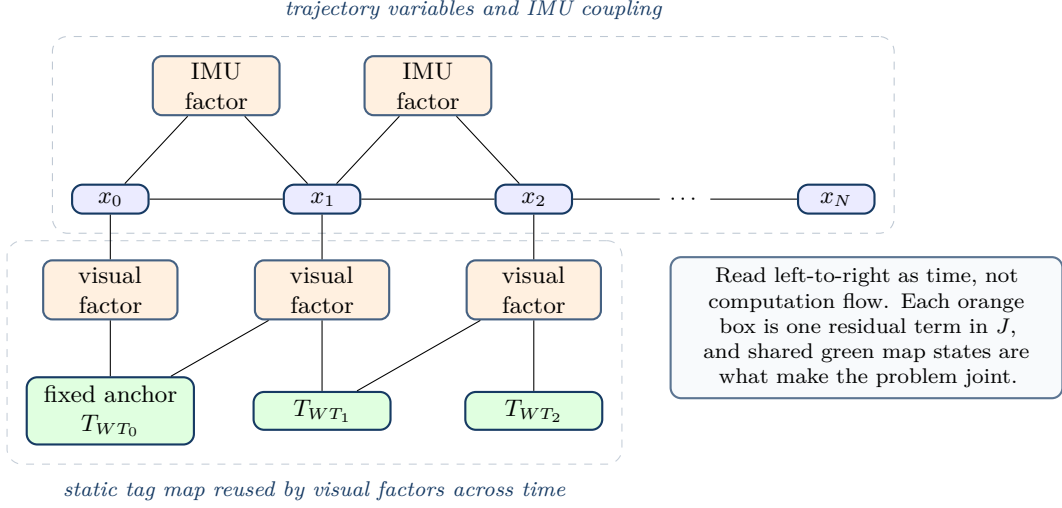


Figure 1.3: Sparsity pattern of the exact joint MAP problem. Body states x_k and static tag poses T_{WT_j} are optimized together; IMU factors couple consecutive states, while visual factors couple each frame state to the anchor or auxiliary tags visible in that frame. The visual-correlation variables c and b_{map} are omitted for readability.

and then solves

$$(X^*, M^*, c^*, b_{\text{map}}^*) = \text{argmin } J. \quad (1.79)$$

Here J_{prior} collects the negative log prior terms in the chosen local coordinates. This becomes the factor-graph / bundle-adjustment view used by fixed-lag smoothers and modern VIO systems [8, 13, 14]. Figure 1.3 should be read as the sparsity pattern of that exact joint MAP problem after discretizing the trajectory: blue nodes are unknown body states, green nodes are static tag poses, orange boxes are residual factors, and an edge means that residual term depends on that variable. It is therefore a coupling graph, not a process-flow diagram.

The key structural point is the repeated reuse of the same static tag variables. The auxiliary-tag states $T_{WT_1}, T_{WT_2}, \dots$ connect to visual factors from multiple frames, so the exact posterior is genuinely joint over trajectory and map. A modular camera-message architecture later in the report is therefore a deliberate approximation to this graph, not the graph itself.

Route C: approximate the filtering posterior recursively by a Gaussian. One computes only the sequence of filtering posteriors of a reduced Markov state and represents each by a local Gaussian. In the present report, the preferred recursive route is either a joint raw-residual EKF update or a message-based information update; the pose-summary EKF is retained only as a weaker fallback. In all cases one is approximating the same filtering posterior

$$p(\chi_k, M \mid \mathcal{D}_{0:k}), \quad (1.80)$$

replacing it by a Gaussian in local coordinates around a nominal state. This yields EKF, IEKF, or related recursive filters [2, 7, 15].

1.6.2 From the continuous posterior to the discrete Bayesian recursion

New notation	Meaning
χ_n	Markov state used by the reduced event-level model, typically $\chi_n = (x_n, c_n, b_{\text{map}})$ or an augmentation of it.
f_n	Event-time transition kernel induced by inertial dynamics and any retained nuisance-state dynamics.
λ_k	Prior-free camera message acting on a local subset of the state and map.
z_k^v	Pose pseudo-measurement used only in the weakest reduced model.

To obtain an online estimator, sample the path at event times and build a finite-dimensional Markov state. The extended sampled prior from (1.43) is Markov on $\tilde{x}_n = (R_n, p_n, v_n, \omega_n, b_{g,n}, b_{a,n})$. Practical filters usually condition on the IMU packets and retain the reduced body state

$$x_n = (R_n, p_n, v_n, b_{g,n}, b_{a,n}).$$

If the colored visual error c_n and the static bias b_{map} are estimated explicitly, define $\chi_n = (x_n, c_n, b_{\text{map}})$; otherwise they are omitted from χ_n and absorbed into an inflated visual model. The event-driven recursion then has two cases.

IMU event. If $s_n = \text{imu}$, propagate from event $n - 1$ to event n :

$$p^-(\chi_n, M) = \int f_n(\chi_n | \chi_{n-1}, u_n) p(\chi_{n-1}, M | \mathcal{D}_{1:n-1}) d\chi_{n-1}. \quad (1.81)$$

Camera event. If $s_n = \text{cam}$ and the corresponding frame index is k , then either update directly with the pixel likelihood or, in the preferred modular approximation, with the message λ_k :

$$p(\chi_n, M | \mathcal{D}_{1:n}) \propto \lambda_k(\chi_n, M) p^-(\chi_n, M). \quad (1.82)$$

Only in the weakest reduction does one replace λ_k by a pseudo-measurement kernel $\tilde{g}_k(z_k^v | \chi_n; \widehat{M}_k)$. Everything that comes later—particle filters, MAP smoothers, EKF, IEKF—is an approximation either to this event-driven recursion or, one level up, to the original raw-pixel posterior measure.

1.6.3 How the Kalman filter is obtained as a simplification

New notation	Meaning
$\hat{X}_n, P_n$	Nominal joint state and covariance at event time τ_n .
δX_n	Local error coordinates for the joint dynamic-plus-map state.
F_n, G_n, H_k	Jacobians of the transition, noise injection, and direct visual residual models.
Λ_n, η_n	Information matrix and information vector of the Gaussian approximation.
A_k	Selector matrix that injects the local camera message into the joint state.

The Kalman filter is exact only for linear dynamics, linear measurements, and Gaussian noise in a vector space. Our problem violates all three conditions: the process and measurement models are nonlinear, and orientation / pose live on manifolds. The practical route to a Kalman-like algorithm is therefore:

1. choose a discrete event-time state sampled from the continuous curve;
2. propagate a nominal state by the nonlinear inertial model through each IMU event;
3. define a local Euclidean error state in tangent coordinates;
4. linearize the transition and camera-assimilation operators around the current nominal state;
5. approximate the local filtering posterior by a Gaussian in the error coordinates.

This is the error-state EKF / IEKF construction.

For the present problem the nominal dynamic state is

$$\hat{x}_n = (\hat{R}_n, \hat{p}_n, \hat{v}_n, \hat{b}_{g,n}, \hat{b}_{a,n}),$$

and a joint filter augments it by local errors for the actively estimated tag poses. The local error vector is therefore

$$\delta X_n = \begin{bmatrix} \delta\theta_n \\ \delta p_n \\ \delta v_n \\ \delta b_{g,n} \\ \delta b_{a,n} \\ \delta m_{1,n} \\ \vdots \\ \delta m_{J,n} \end{bmatrix} \in \mathbb{R}^{15+6J}. \quad (1.83)$$

If quaternions are used internally, $\delta\theta_n$ is still 3-dimensional; only the stored nominal orientation changes from $\hat{R}_n$ to $\hat{q}_n$. If c_n and/or b_{map} are estimated explicitly, then δX_n is augmented by the corresponding Euclidean error coordinates.

For an IMU event, the linearized prediction step is

$$\delta X_n = F_n \delta X_{n-1} + G_n w_n, \quad P_n^- = F_n P_{n-1}^+ F_n^\top + G_n Q_n G_n^\top. \quad (1.84)$$

If the static map is part of the same state, its nominal value does not move, but the cross-covariances with the dynamic state do propagate. This is exactly the statistical dependence that a separated pose-summary architecture tends to lose.

At a camera event there are three implementation levels:

1. **Direct raw-residual EKF update.** Linearize the stacked pixel residual itself,

$$r_k \approx H_k \delta X_n + \nu_k, \quad \nu_k \sim \mathcal{N}(0, R_k),$$

and apply the ordinary Joseph-form update

$$S_k = H_k P_n^- H_k^\top + R_k, \quad K_k = P_n^- H_k^\top S_k^{-1}, \quad P_n^+ = (I - K_k H_k) P_n^- (I - K_k H_k)^\top + K_k R_k K_k^\top. \quad (1.85)$$

This is the most direct recursive approximation to the pixel likelihood.

2. **Preferred modular information update.** If the camera inversion returns a prior-free message on a local variable block, inject it into the joint Gaussian in information form:

$$\Lambda_n^+ = \Lambda_n^- + A_k^\top \Delta \Lambda_k A_k, \quad \eta_n^+ = \eta_n^- + A_k^\top \Delta \eta_k, \quad (1.86)$$

where A_k selects the current camera-pose and visible-tag coordinates touched by the frame. One then converts back to covariance form if needed. This realizes the event-driven KF while preserving the meaning of the camera contribution as a likelihood factor rather than as a recycled posterior.

3. **Fallback pose-summary update.** If one keeps only

$$h_k(x_k, c_k, b_{\text{map}}) = \text{Log}(\hat{T}_{WC,k}^{-1} T_{WB,k} T_{BC}) - b_{\text{map}} - c_k, \quad (1.87)$$

together with covariance $R_{v,k}$, then the update is the familiar pose pseudo-measurement EKF. This is the easiest implementation, but it is lower fidelity and must not be built from a prior-conditioned camera posterior unless that prior has been removed first.

Finally, the corrected local mean is injected back onto the manifold by the same retraction rules used elsewhere in the report:

$$\hat{R}_n^+ = \hat{R}_n^- \text{Exp}([\delta \hat{\theta}_n]_\times), \quad \hat{p}_n^+ = \hat{p}_n^- + \delta \hat{p}_n,$$

with analogous additive updates for velocity, biases, and any explicitly retained Euclidean nuisance states. Thus the Kalman filter is not a different model from the Bayesian inverse problem. It is a tractable local-Gaussian approximation to the same event-driven posterior, obtained after discretization and linearization.

1.6.4 When the Kalman simplification is appropriate and when it is not

New notation	Meaning
Unimodal posterior	A posterior well approximated by one Gaussian bump in local coordinates.
Linearization basin	Region in which the current nominal state is close enough to the true state for linearization to be accurate.
Reacquisition	First visual correction after a visual dropout interval.

The EKF / IEKF approximation is appropriate when the posterior remains close to unimodal, the visual updates arrive frequently enough to keep inertial drift under control, and initialization starts inside the basin of attraction of the correct solution. Within that regime, the direct raw-residual update and the prior-free message update are both defensible. The pose-summary fallback is weaker, but can still be serviceable if its covariance is honest and its approximation error is monitored.

The approximation becomes weak when long visual dropouts allow inertial propagation to drift far from the truth, when the camera inversion becomes ambiguous or produces multimodal hypotheses, when outliers are not sufficiently gated, or when the visual approximation is badly misspecified. In such regimes a fixed-lag smoother or a more robust sampling-based approximation is safer. This is why the practical implementation hierarchy later in the report recommends starting from the message-aware event-driven filter and then moving to fixed-lag smoothing when relinearization, delayed correction, or correlation repair becomes important.

Chapter 2

Practical Implementation

Roadmap. This chapter translates the model into implementation choices. It starts with parameter roles and representative hardware scales, then specifies the event-driven architecture, initialization and tag bookkeeping, and finally lists approximation risks and validation statistics.

2.1 Parameter and noise guide: prior, likelihood, and tuning roles

New notation	Meaning
m_0, P_0	Mean and covariance of the initial-state prior.
Q_v, Q_ω	Trajectory-smoothness spectral densities in the prior-on-curves.
Q_{bg}, Q_{ba}	Bias random-walk spectral densities in the prior.
R_g, R_a	IMU likelihood covariance or equivalent noise-density parameters.
$\Delta\Lambda_k, \Delta\eta_k$	Information-form strength and shift of the camera message at frame k .
$R_{v,k}$	Fallback pose-summary covariance when a frame is collapsed to z_k^v .
$\sigma_q, \sigma_0, \alpha_A, \alpha_\theta, \alpha_b$	Pixel-to-Hessian or pixel-to-pose noise parameters used to build the visual approximation.
ρ_p, ρ_R, Q_c	Colored visual-noise hyperparameters for the latent process c_k .
Σ_{map}	Prior covariance of the sequence-level map / anchor bias term.

A frequent source of confusion is mixing up *prior parameters* and *likelihood parameters*. The matrices $Q_v, Q_\omega, Q_{bg}, Q_{ba}$ and the initial parameters m_0, P_0 define the prior law. They say what kinds of trajectories and bias evolutions are plausible *before* the current measurements are assimilated. By contrast, R_g, R_a , the frame-wise camera Hessian, and any derived quantities such as $(\Delta\Lambda_k, \Delta\eta_k)$ or the fallback covariance $R_{v,k}$ describe the spread and information content of the observed measurements around the model-predicted values.

A helpful mental model is:

- prior parameters regularize the *unknown trajectory*;
- likelihood parameters regularize the *mismatch between data and model*;
- nuisance-state priors, such as Q_c or Σ_{map} , sit in between and regularize latent error processes that explain persistent discrepancies.

In practice, parameter selection should proceed hierarchically.

1. Estimate R_g, R_a and their continuous-time equivalents from stationary IMU data, preferably through Allan deviation analysis. Kalibr’s IMU model uses continuous-time white-noise and bias-random-walk parameters of exactly this form [6].
2. Estimate or model the camera-message strength frame by frame from the local visual Hessian or from a parametric geometry-driven model such as (1.70); if a pose-summary fallback is used, this same step yields $R_{v,k}$.
3. Set m_0 from the first trustworthy anchored visual pose and P_0 by inflating the corresponding local covariance to reflect unmodeled setup uncertainty.
4. Tune $Q_v, Q_\omega, Q_{bg}, Q_{ba}$ so that innovation statistics are consistent: too small gives under-dispersed posteriors, too large gives noisy or drifting estimates.

5. If residual autocorrelation remains high on the visual side, introduce c_k with parameters (ρ_p, ρ_R, Q_c) rather than forcing independent noise to explain correlated errors.
6. If the camera module internally uses a predicted prior, verify that the exported visual interface is prior-free by construction; otherwise the back-end will be systematically overconfident.

2.2 Representative numeric regime for a phone-on-arm lab experiment

New notation	Meaning
Δt_c	Camera frame interval.
f_{imu}	IMU sampling rate.
s	Tag side length.
z	Camera-to-anchor range.
$\sigma_g, \sigma_a, \sigma_{bg}, \sigma_{ba}$	IMU white-noise and bias-random-walk strengths.

This section gives two kinds of numbers: *hardware facts* drawn from cited sources, and *engineering starting values* for the model parameters used in this chapter. The latter are recommendations, not immutable truths.

Quantity	Representative value	Rationale / source
Phone video rate	60 fps	Google’s dedicated Pixel 9a specs page lists rear-camera 4K and 1080p recording at 30/60 fps; taking the 60 fps mode as the working regime gives $\Delta t_c \approx 16.7$ ms for this example [16].
Phone main-camera FoV	about 82°	Google’s Pixel hardware tech-specs page lists an 82° field of view for the Pixel 9a wide camera [16].
Phone motion-sensor rate	200 Hz limit for <code>registerListener()</code> on Android 12+ apps without the high-rate permission	Android documents that, for apps targeting Android 12 (API level 31) or higher, <code>registerListener()</code> is rate-limited to 200 Hz for the affected motion and position sensors unless the app declares <code>HIGH_SAMPLING_RATE_SENSORS</code> [17].
Research tabletop arm reach	about 0.855 m	The Franka Research 3 product page lists an 855 mm reach; this is a realistic scale for anchor-to-workspace geometry in a lab manipulation setup [18].
Robot repeatability	$< \pm 0.1$ mm	The official Franka Research 3 datasheet lists position repeatability below ± 0.1 mm under ISO 9283 [18].
Robot control rate	about 1 kHz low-level control	Official FR3 materials describe 1 kHz control and measurement access through the Franka Control Interface; this is much faster than 60 Hz phone video and faster than the 200 Hz Android sensor-rate limit without the high-rate permission [16–18].
Practical tag sizes	0.06–0.10 m	Typical lab fiducials for arm-mounted phone experiments are in the 6–10 cm range; this is an engineering choice that keeps tags visible at about 0.4–1.2 m range.
Practical anchored range	0.4–1.2 m	Consistent with a tabletop arm of roughly 0.85 m reach plus camera offset and the need to keep the anchor visible at least at startup.
Example Kalibr <code>imu.yaml</code> values	$\sigma_a = 1.86 \times 10^{-3}$ m/s ² /√Hz, $\sigma_{ba} = 4.33 \times 10^{-4}$ m/s ³ /√Hz, $\sigma_g = 1.87 \times 10^{-4}$ rad/s/√Hz, $\sigma_{bg} = 2.66 \times 10^{-5}$ rad/s ² /√Hz	Kalibr’s YAML-formats page shows these as example <code>imu.yaml</code> entries at an update rate of 200 Hz; the units and continuous-time interpretation are documented separately in the Kalibr IMU-noise page, so these values are useful as a format-and-units example, not as a phone-specific default [6, 19].

Table 2.1: Representative numbers for a phone-on-arm laboratory regime. Some are hardware facts from official specs; others are engineering starting points for model design or sanity-check performance levels.

For a first implementation aimed at a robot arm moving in a lab around an anchored tag, a very reasonable working regime is:

- camera at 60 Hz,
- IMU at 100–200 Hz,
- tag size $s \in [0.06, 0.10]$ m,
- anchor range $z \in [0.4, 1.2]$ m,
- path spans on the order of 0.1–0.5 m in translation over the sequence,
- translational speeds on the order of 0.05–0.5 m/s for careful tabletop motions,

- angular rates on the order of $5^\circ/\text{s}$ to $60^\circ/\text{s}$ for ordinary camera reorientation.

These last values are engineering design targets rather than universal physical constants; they simply place the prior and likelihood parameters in the right numerical regime for a lab arm rather than a UAV or car.

2.3 Implementation architecture for the anchored multi-tag problem

New notation	Meaning
$M_k = \{(\hat{T}_{WT_j,k}, P_{m_j,k})\}_{j \in \mathcal{J}_k}$	Current map manager state: a set of auxiliary-tag pose posteriors carried up to frame k .
$P_{m_j,k} \in \mathbb{R}^{6 \times 6}$	Local covariance of tag j in tangent coordinates around $\hat{T}_{WT_j,k} \in \text{SE}(3)$.
$\mathcal{J}_k$	Set of auxiliary tags that have already been initialized by frame k .
$\mathcal{V}_k^{\text{old}}, \mathcal{V}_k^{\text{new}}$	Visible tags at frame k that are already in the map, and visible tags observed for the first time.
λ_k or $(\Delta\Lambda_k, \Delta\eta_k)$	Prior-free visual message returned by the camera inversion at frame k .
$z_k^v = (\hat{T}_{WC,k}, R_{v,k})$	Weaker fallback pose summary retained for simplified implementations.

The previous chapter established the reference posterior and the message-based reduced interface. This section translates those objects into an implementation architecture. The central design choice is now:

- If accuracy and statistical correctness are the highest priorities, solve a *joint* posterior over trajectory and static tag map.
- If online speed and modularity are the highest priorities, use an *event-driven message architecture*: the camera module may solve its own local subproblem, but it must export a prior-free visual message to the joint recursive estimator.
- Use a pose pseudo-measurement z_k^v only as a weaker fallback when one explicitly accepts the extra loss of correlation structure.

The rest of the section explains how that architectural choice affects initialization, new-tag birth, reappearance handling, and map bookkeeping. Figure 2.1 should be read as an engineering module diagram for the event-driven modular approximation, not as a posterior graph like Figure 1.3. Solid arrows denote current-event information flow, while the dashed path denotes persistent map memory and cached priors reused by later camera events.

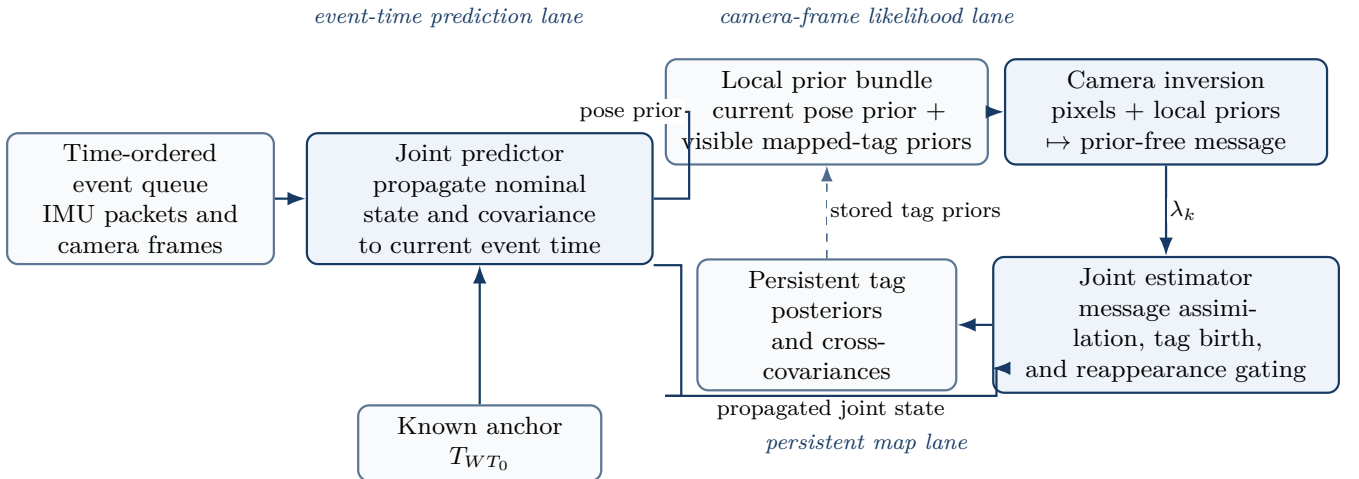


Figure 2.1: Event-driven implementation architecture. IMU and camera events are processed in timestamp order. The predictor advances the joint state to the current event time; camera frames trigger a local inversion that returns a prior-free message; the joint estimator assimilates that message while preserving pose-map cross-covariances. Persistent tag beliefs are stored and reused by later camera events.

2.3.1 The exact joint solution

The most defensible solution is to keep the entire unknown in one posterior. In discrete time this means solving for

$$X_{0:N} = \{x_0, \dots, x_N\}, \quad M = \{T_{WT_j}\}_{j \in \mathcal{J}_{\text{aux}}},$$

with posterior density

$$p(X_{0:N}, M \mid \mathcal{U}_{0:N-1}, Y_{0:N}) \propto p_0(x_0)p_0(M) \prod_{k=0}^{N-1} p(x_{k+1} \mid x_k, \mathcal{U}_k) \prod_{k=0}^N p(Y_k \mid x_k, M).$$

No information is used twice: each IMU packet contributes once through the process factor, each corner pixel contributes once through the camera model, and the static map and trajectory are inferred together. In practice this posterior is solved either by fixed-lag smoothing, batch MAP estimation, or posterior sampling as discussed earlier.

The fully joint method is especially attractive when

- anchor visibility is intermittent,
- the auxiliary tags are numerous enough to act as a map,
- one wants statistically honest uncertainty on both camera pose and tag pose,
- and the same data must later support a trajectory posterior, a map posterior, and control diagnostics.

The price is computation and implementation complexity. A real-time version is usually built as a fixed-lag smoother, an EKF-SLAM style filter, or an incremental factor-graph system with a bounded set of actively tracked tags [7, 8, 13, 20].

2.3.2 The recommended event-driven message architecture

The recommended real-time alternative is to keep the event-driven timeline but modularize the camera processing. The estimator is then separated into three coupled pieces:

1. a joint recursive estimator that propagates the dynamic state and the active tag-map covariance through IMU events,
2. a camera inversion layer that uses the current local prior and the current pixels to form a camera-only or pose-plus-visible-tag posterior,
3. a message interface that removes the local prior and feeds only the visual likelihood contribution back into the joint estimator.

For IMU events, the joint state is propagated in the usual way. For camera events, the recursion is

$$p_k^-(x_k, M) = p(x(t_k), M \mid \mathcal{D}_{<k}), \quad (2.1)$$

$$q_k^+(z_k) \approx p(Y_k \mid z_k) q_k^-(z_k), \quad (2.2)$$

$$\lambda_k(z_k) \propto q_k^+(z_k)/q_k^-(z_k), \quad (2.3)$$

$$p_k^+(x_k, M) \propto \lambda_k(x_k, M) p_k^-(x_k, M), \quad (2.4)$$

where the quotient is understood in the local Gaussian or local Laplace sense described in Chapter 1. This architecture preserves the main Bayesian rule: the same prior is not reused twice.

This design is still approximate, because the camera inversion itself is local and usually nonlinear, and because only a subset of the full joint posterior is exposed to the camera block at each frame. It is nevertheless more faithful to the Bayesian factorization than a pose-summary-only pipeline. A clean implementation should obey the following rules.

- IMU packets should enter the estimator exactly once, through event-time propagation.
- If the camera inversion uses the predicted pose and tag priors, it must export a prior-free message rather than its posterior.
- Newly seen tags should be born with broad covariance and should not strongly correct global pose on first sighting.
- Reappearing tags should be gated before they are trusted, because stale or moved tags can damage both pose and map.
- Any slower background map refinement or fixed-lag smoother should be described as a posterior-improvement layer, not as a second independent sensor update.

If the message layer is unavailable, a pose pseudo-measurement update remains possible, but it should be labeled honestly as a weaker approximation. That fallback discards the most important camera–map couplings precisely where multi-tag localization is strongest.

Figure 2.2 makes the exposure-time logic explicit. The estimator does not wait for an artificial camera–IMU pair; it propagates through the arriving IMU packets and, when a camera frame arrives, advances the state to that exact exposure time before launching the camera inversion.

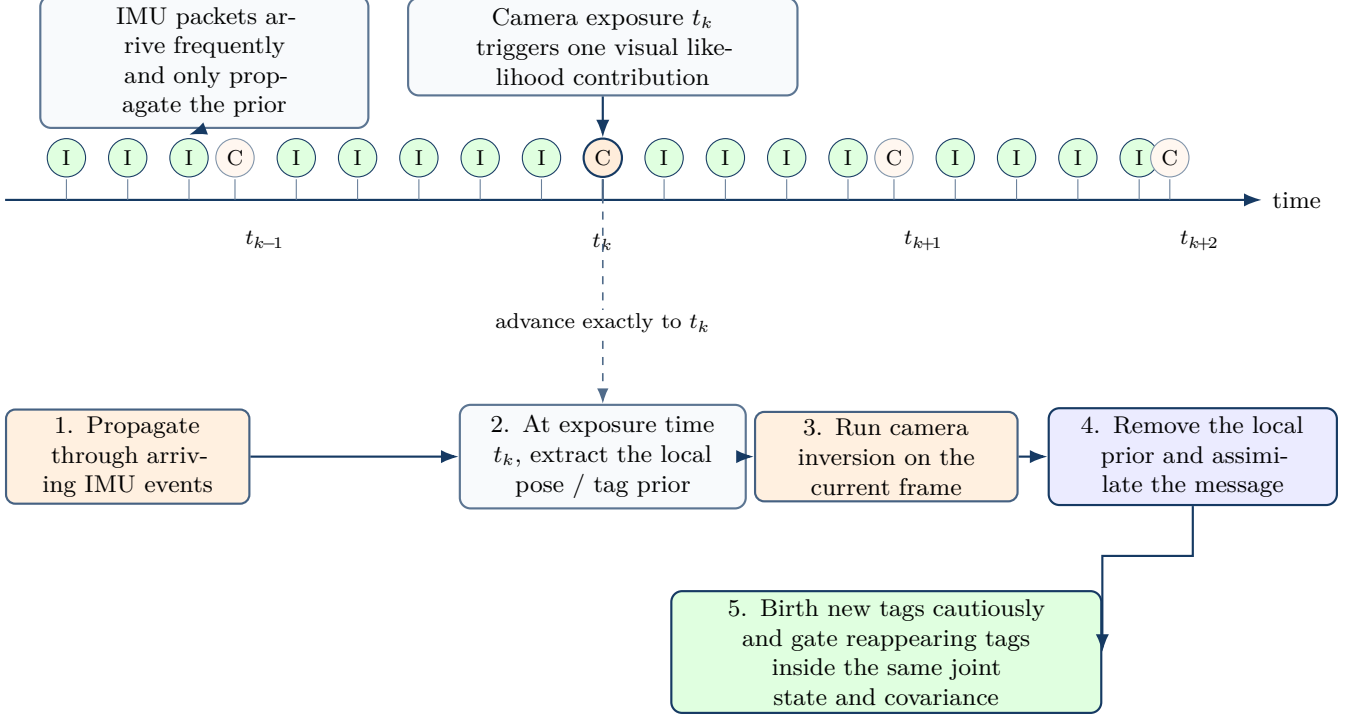


Figure 2.2: Exposure-time flow in the recommended modular approximation. Top: asynchronous IMU and camera events are processed strictly in timestamp order. Bottom: when a camera frame arrives at exposure time t_k , the estimator first advances to that exact time, extracts the local prior, runs the camera inversion, removes the prior contribution, and assimilates the resulting visual message into the same joint state that stores the active tag posteriors.

2.3.3 First frame: anchor guaranteed visible

At frame $k = 0$ the anchor is assumed visible. This makes initialization much simpler because the global gauge is fixed immediately.

Step 1: initialize the camera pose from the anchor. Solve the camera inversion using the anchor tag alone, or the anchor plus any already-known tags if such a map exists, to obtain

$$q(T_{WC,0}) \approx \mathcal{N}_{\mathfrak{se}(3)}(\hat{T}_{WC,0}, R_{v,0}).$$

Then convert to body pose using the known camera–IMU extrinsic, with $T_{CB} = T_{BC}^{-1}$,

$$\hat{T}_{WB,0} = \hat{T}_{WC,0} T_{CB},$$

and propagate covariance through the local linearization.

Step 2: initialize non-pose Euclidean components. Choose

$$v_{W,0} \sim \mathcal{N}(0, P_{v0}), \quad b_{g,0} \sim \mathcal{N}(0, P_{bg0}), \quad b_{a,0} \sim \mathcal{N}(0, P_{ba0}),$$

with conservative spreads unless a stationary segment is available.

Step 3: initialize every auxiliary tag that is already visible. For each $j \in \mathcal{V}_0 \setminus \{0\}$, condition on the current anchored camera belief and solve a tag-pose inverse problem

$$\hat{T}_{WT_j,0} = \operatorname{argmin}_{T \in \mathfrak{SE}(3)} \sum_{i=1}^4 \left\| \tilde{u}_{i,0}^{(j)} - \pi(K, \hat{T}_{CW,0} T P_i^{(j)}) \right\|_{\Sigma_{0,j,i}^{-1}}^2. \quad (2.5)$$

Then initialize a broad covariance $P_{m_j,0}$ from the local Hessian, inflated to reflect uncertainty in the camera pose used for conditioning. Intuitively, the anchor gives the first trustworthy camera pose, and that camera pose gives the first absolute estimate of every co-visible auxiliary tag. These newborn tags should be marked provisional and should not dominate global pose correction until they have been reobserved.

2.3.4 First sighting of a new tag at a later frame (tag birth)

Suppose tag j is seen for the first time at frame $k > 0$. Then there are two legitimate routes.

Joint route. In a joint filter or smoother, augment the state with a new static variable T_{WT_j} and update the enlarged posterior directly using the current pixels. This is the statistically clean route.

Message-layer route. In the recommended modular design, use the current body and visible-tag posterior as a conditional prior and create a new tag posterior

$$p(T_{WT_j} | Y_k^j, \mathcal{D}_{0:k}) \propto \int p(Y_k^j | x_k, T_{WT_j}) p(x_k | \mathcal{D}_{0:k}) dx_k p_0(T_{WT_j}). \quad (2.6)$$

This integral is usually approximated by conditioning on the current body mean or sigma points. A practical implementation is:

1. wait until the current camera/body posterior is well constrained by the anchor or by already-mapped tags,
2. solve (2.5) at the current frame using the current anchored camera mean,
3. assign a broad covariance that includes both pixel uncertainty and the uncertainty in the current camera pose,
4. mark the tag as “provisional” until it has been observed repeatedly.

Important edge case. If the camera sees only entirely new unknown tags and no anchor or already-initialized tags, then those tags cannot yet be anchored absolutely in the world frame. In that case the correct action is *not* to let them strongly correct global pose. One should defer firm birth until a frame where the camera itself is well localized in the anchored frame.

2.3.5 Tag disappearance and later reappearance

A stationary tag that disappears for several frames does not require a special dynamic model. Its pose remains static:

$$T_{WT_j,k+1} = T_{WT_j,k},$$

possibly with a very small artificial process noise added only for numerical conditioning. Therefore during absence one simply carries its posterior forward,

$$q_{k+1}(T_{WT_j}) = q_k(T_{WT_j}),$$

while recording bookkeeping such as last seen time, number of successful updates, and recent residual quality. When the tag reappears, one treats the previous tag posterior as the prior and performs an ordinary measurement update. In the joint formulation, the pixels for that tag simply re-enter the global likelihood. In the message-layer formulation, the current body posterior and previous tag posterior produce a predicted pixel distribution, which is then used for gating and update. Formally,

$$p(T_{WT_j} | Y_k^j, \mathcal{D}_{0:k}) \propto p(Y_k^j | x_k, T_{WT_j}) p(T_{WT_j} | \mathcal{D}_{0:k-1}), \quad (2.7)$$

with the current body posterior either conditioned on jointly or approximated by its mean/covariance. For robust practice, reappearing tags should be accepted only after a consistency gate:

- predicted-vs-observed corner residual below threshold,
- correct decoded tag id,
- plausible depth and viewing geometry,
- and, if available, Mahalanobis innovation or NIS below a chosen threshold.

If a tag has been absent for a long interval, mild covariance inflation before the update is a defensible safeguard against overconfidence.

2.3.6 How to keep track of all tags in code

A simple and effective representation is to store each auxiliary tag as a record

$$m_j = (\hat{T}_{WT_j}, P_{m_j}, n_j, \tau_j, \eta_j),$$

where n_j is observation count, τ_j is last-seen timestamp, and η_j is a small quality bundle such as mean residual, provisional/confirmed flag, and recent visibility streak. The whole map manager state is then

$$M_k = \{m_j\}_{j \in \mathcal{J}_k}.$$

No global alignment other than the anchor is assumed: every tag carries its own arbitrary 6-DoF pose in the anchored world frame. In a joint filter, these records also imply cross-covariance blocks with the dynamic state. In a slightly more modular implementation, dormant tags can be stored out of the active state and activated only when visible, but the same local Lie-coordinate conventions still apply.

The camera pose itself is tracked in exactly the same geometric space as each tag pose, namely a pose on $SE(3)$ plus a covariance on the corresponding tangent space. The difference is semantic, not mathematical: the camera pose is dynamic and time indexed, while each auxiliary-tag pose is static and time invariant.

2.3.7 A compact implementation recipe

For readers who want an executable checklist, the following is a good starting point.

1. **Sort all events by timestamp.** Do not force camera and IMU data into synchronized pairs.
2. **Frame 0.** Use anchor pixels to initialize $T_{WC,0}$, then initialize $T_{WB,0}$, velocity, and biases. Initialize all co-visible auxiliary tags as provisional static poses with broad covariance.
3. **At each IMU event.** Propagate the nominal state and covariance to the event time, preserving cross-covariances with active tags.
4. **At each camera event.** Propagate to the exact exposure time and extract the local prior over current pose and visible known tags.
5. **Run camera inversion.** Use all visible tag pixels and the local priors to build a local posterior.
6. **Remove the local prior.** Export only the prior-free visual message, not the camera posterior itself.
7. **Assimilate jointly.** Update the body state and all visible active tags inside the same covariance or information matrix.
8. **Birth new tags cautiously.** Initialize them with broad covariance and prevent them from strongly correcting global pose until reobserved.
9. **Gate reappearing tags.** Use residual and NIS checks before accepting them into the update.
10. **Bookkeeping.** Increment counts, store residual statistics, mark tags as confirmed after enough successful reobservations, and keep absent tags unchanged until they reappear.

This recipe keeps the module boundaries clean while preserving the key Bayesian rule that each physical frame contributes one visual likelihood factor exactly once.

2.4 Kalman realization, sampling, and approximation pitfalls

New notation	Meaning
$\pi(\cdot)$	Generic posterior density.
θ	Collective variable used in sampling, e.g. a block of poses or tag states.
$\alpha(\theta, \theta')$	Metropolis–Hastings acceptance probability for a proposal $\theta \rightarrow \theta'$.
Λ, η	Information matrix and information vector of the local Gaussian approximation.
NIS, NEES	Innovation and state consistency statistics used to validate the Gaussian approximation.

2.4.1 How the Kalman filter emerges from the Bayesian problem

The exact filtering recursion was written in (1.82). A practical Kalman-style realization appears only after several simplifications:

1. discretize the curve prior into a Markov transition law,
2. represent the posterior by a local Gaussian in tangent coordinates,
3. linearize the process and measurement operators,
4. assume that the necessary marginal or conditional distributions are adequately captured by their first two moments.

The implementation consequence is that the filter should carry the current dynamic state and any active tag states in one Gaussian object. If the covariance is partitioned as

$$P_n^- = \begin{bmatrix} P_{xx,n}^- & P_{xM,n}^- \\ P_{Mx,n}^- & P_{MM,n}^- \end{bmatrix},$$

then an IMU propagation updates not only the dynamic block but also the cross-covariances:

$$\bar{x}_n = f(\hat{x}_{n-1}, u_n), \quad \bar{P}_{xx,n} = F_n P_{xx,n-1} F_n^\top + G_n Q_n G_n^\top, \quad \bar{P}_{xM,n} = F_n P_{xM,n-1}.$$

That line is easy to miss and is exactly why separated tag-map updates tend to become overconfident.

At a camera event, the preferred reduced implementation is the information-form message update from (1.86). One keeps

$$\Lambda_n^- = (P_n^-)^{-1}, \quad \eta_n^- = \Lambda_n^- \mu_n^-,$$

adds the message, and converts back to covariance form only if needed. This is the best modular compromise: the camera code stays self-contained, but the estimator still treats the frame as one likelihood contribution. The raw-residual EKF update is formally cleaner and should be preferred whenever the codebase can tolerate direct camera residuals inside the estimator. The pose pseudo-measurement update is the easiest engineering shortcut, but it should be treated as a downgrade in statistical fidelity rather than as the canonical Kalman realization.

2.4.2 Metropolis–Hastings sampling for posterior visualization

For scientific understanding and posterior visualization, it is useful to sample not only trajectories but also tag poses. Let

$$\theta = (X_{0:N}, M)$$

or, for a fixed-lag window, a smaller block such as recent poses, recent biases, and the currently visible tag poses. Given a proposal density $q(\theta' | \theta)$, the Metropolis–Hastings acceptance probability is

$$\alpha(\theta, \theta') = \min \left\{ 1, \frac{\pi(\theta') q(\theta | \theta')}{\pi(\theta) q(\theta' | \theta)} \right\}. \quad (2.8)$$

A practical choice is blockwise proposals in local coordinates:

- propose several consecutive body poses in $\mathfrak{se}(3)$ coordinates,
- propose velocity and bias blocks additively in $\mathbb{R}^3$,
- propose one or more tag poses in $\mathfrak{se}(3)$ coordinates.

This is not meant for real-time estimation at 60 Hz. Its purpose is diagnostic and pedagogical: it reveals multimodality, non-Gaussianity, and tag–trajectory coupling that a local EKF or message-linearization approximation may hide. For example, one can visualize posterior samples of a reappearing tag’s pose, or posterior tubes of the camera trajectory during partial occlusion.

2.4.3 Common probabilistic failure modes

Several implementation shortcuts are acceptable approximations. Several others break the intended posterior semantics and should be treated as modeling errors.

- **Reusing a camera posterior as a measurement.** If the camera inversion uses the predicted pose prior and its posterior is fed back as if it were an independent sensor reading, the prior has been counted twice.
- **Double counting IMU.** If IMU is used inside the camera module to form a visual summary and then used again in the back-end process model, inertial information has been counted twice.
- **Using the same pixels twice as if independent.** If a frame’s pixels are first condensed into a posterior and then used again to update the same latent variables without accounting for dependence, posterior uncertainty becomes too small.
- **Birth of a new tag without an anchored camera posterior.** A tag seen in isolation cannot be assigned a trustworthy absolute world pose unless the camera pose in the anchored frame is already known.
- **Treating tag poses as Euclidean vectors.** Tag pose covariance must live in local coordinates on $SE(3)$, not in arbitrary matrix-entry coordinates.
- **Ignoring time correlation in reduced visual summaries.** If the front-end reuses similar geometry frame after frame, assuming independent $R_{v,k}$ may lead to overconfident trajectory estimates.

On the other hand, the following are usually acceptable first approximations:

- using the current camera mean to initialize a new tag and inflating its covariance,
- keeping a slower background map-refinement module or fixed-lag smoother for dormant tags,
- using a fixed-lag smoother to repair the neglected correlations of the real-time front-end.

2.4.4 Assessment of the modular message scheme

The modular scheme is defensible when it is described as an approximation to the joint posterior, not as the joint posterior itself. The recommended reduced architecture is the event-driven message architecture described above. The camera code can stay separate, the back-end can remain recursive and real-time, and a slower map-refinement layer can still be added later. The key requirement is that a prior-conditioned camera posterior must not be treated as an independent sensor measurement unless the prior contribution has first been removed.

The direct raw-residual joint filter remains formally cleaner whenever it is computationally feasible. The pose-summary fallback remains usable for a first prototype or ablation study, but it should be documented as a weaker approximation whose neglected couplings and prior-reuse hazards can make the filter overconfident. This places the practical hierarchy in a clear order: exact joint posterior first, message-based recursive approximation second, pose-summary shortcut last [7, 8, 13, 14, 20].

2.5 Synthesis and literature position

The reference problem is a posterior on a trajectory curve together with a static auxiliary-tag map in an anchored world frame. The prior is a law on that mixed object: smooth body motion, slowly drifting biases, and static but initially unknown tag poses. In the full joint model the IMU contributes inertial transition likelihoods and the camera contributes the raw-pixel likelihood through the tag projection model. If one solves everything jointly, the result is a direct Bayesian estimator over both trajectory and map. If one instead chooses a speed-friendly decomposition, the defensible reduced model is the event-driven message architecture: IMU packets are used exactly once in propagation, the camera module converts each frame into a prior-free visual message, and the joint recursive estimator assimilates that message while preserving pose-map cross-covariances for the active states. Pose pseudo-measurements remain only as a weaker fallback. The practical message is therefore simple: exact Bayes is the reference model, and recursive or fixed-lag estimators are computational approximations to it.

2.5.1 How the literature positions these choices

New notation	Meaning
Markov prior	Prior that factorizes as $p(x_0) \prod_k p(x_{k+1} x_k)$.
GP prior	Gaussian-process prior on the whole continuous-time trajectory curve.
MAP	Maximum a posteriori estimate: the minimizer of the negative log posterior.
Fixed lag	Approximate smoothing over a moving recent time window.

The Bayesian inverse-problem perspective treats trajectory and map as latent variables inferred from sensor data and priors [4]. Recursive Bayesian filtering and smoothing provide the general algorithmic backbone [15]. In visual-inertial robotics, filtering families such as MSCKF keep a Gaussian posterior over the current state and selected pose history [7]. Optimization families such as IMU-preintegrated MAP smoothing, OKVIS, and VINS-Mono instead solve nonlinear MAP problems over keyframes or sliding windows of recent states, sometimes with structureless visual factors rather than explicit landmark states [8, 13, 21, 22]. Fiducial-tag systems such as TagSLAM explicitly represent static tag poses and body poses together in a factor graph [20]. Continuous-time Bayesian approaches such as STEAM place priors directly on the space of trajectories and then condition on observations [23, 24]. The anchored multi-tag method developed in this report fits naturally in this landscape as either a joint fiducial visual-inertial SLAM estimator or a conservative event-driven recursive approximation with a camera-message layer.

Representative work	Unknown object	Prior / regularizer	Inference style	What it assumes or emphasizes
Stuart (2010) [4]	Function or curve in infinite-dimensional space	Prior measure on the whole unknown, often Gaussian on a function space	Full Bayesian posterior on the unknown space	General Bayesian inverse viewpoint; emphasizes posterior well-posedness and priors on spaces of functions.
Apte et al. (2007) [5]	Continuous-time path	Path-space prior induced by model dynamics and noise	MCMC / path-space Bayesian assimilation	Explicitly formulates data assimilation as conditioning a probability law on paths.
Arulampalam et al. (2002) [25]	Discrete state trajectory	Markov state-transition prior	Recursive Bayesian filtering, including particle filters	Generic nonlinear / non-Gaussian online Bayesian tracking.
Mourikis–Roumeliotis (2007) [7]	Pose, velocity, biases, camera clones	IMU-driven Gaussian Markov prior with bias random walks	EKF / MSCKF	Real-time vision-aided inertial navigation with recursive updates.
Leutenegger et al. (2015) [13]	Keyframe poses, velocities, biases, landmarks	Gaussian priors plus IMU factors and robust visual residuals	Nonlinear MAP / fixed-lag optimization	Tightly coupled VIO with sliding-window refinement.
Forster et al. (2017) [8]	Keyframe states plus preintegrated increments	Discrete Gaussian motion prior expressed by on-manifold IMU factors	MAP / factor graph / incremental smoothing	Efficient manifold-correct inertial factors and delayed bias correction.
Anderson–Barfoot–Tong–Särkkä (2015) [24]	Entire continuous-time trajectory curve	GP prior generated by a linear time-varying SDE	Batch GP regression / smoothing	Continuous-time prior, interpolation at arbitrary times, exact sparsity.

Table 2.2: Representative Bayesian formulations for trajectory-pose estimation. The same inverse problem can be expressed through path-space priors, discrete Markov priors, MAP objectives, or continuous-time Gaussian-process priors. The table is a positioning aid rather than an exhaustive survey.

Appendix A

Prior samples for interpretation and sanity checking

Roadmap. The appendix visualizes the trajectory prior before any measurements are used. It is a sanity check for the chosen motion and bias hyperparameters, not a posterior result and not a claim about one unique physical experiment.

New notation	Meaning
Δt	Discrete time step used for sampling; here $\Delta t = \frac{1}{60}$ s.
N	Number of discrete states; in the annex $N = 241$ corresponding to a 4 s horizon.
a_k, α_k	Sampled translational acceleration and angular-acceleration drives at step k .
ω_k	Latent body angular velocity used only to generate the pose process in this annex.
rpy_k	Roll–pitch–yaw angles extracted from R_k only for visualization; they are <i>not</i> the coordinates used by the filter.

This annex does not introduce a new inferential model. It samples a simple discrete-time approximation to the extended trajectory prior from Chapter 1. The purpose is pedagogical: it is much easier to understand the role of the prior hyperparameters once one sees typical paths they generate. In particular, the sampler keeps the latent angular velocity ω_k explicit so that the orientation process can be generated directly. This matches the extended-state prior discussion in Chapter 1; it is not literally the same object as the reduced IMU-conditioned filter recursion used later for online estimation.

For the simulation we use the discrete generative scheme

$$p_{k+1} = p_k + v_k \Delta t + \frac{1}{2} a_k \Delta t^2, \quad (\text{A.1})$$

$$v_{k+1} = v_k + a_k \Delta t, \quad (\text{A.2})$$

$$R_{k+1} = R_k \text{Exp}([\omega_k]_{\times} \Delta t), \quad (\text{A.3})$$

$$\omega_{k+1} = \omega_k + \alpha_k \Delta t, \quad (\text{A.4})$$

$$b_{g,k+1} = b_{g,k} + \eta_k^{bg}, \quad (\text{A.5})$$

$$b_{a,k+1} = b_{a,k} + \eta_k^{ba}, \quad (\text{A.6})$$

with independent Gaussian drives

$$a_k \sim \mathcal{N}(0, \sigma_a^2 I_3), \quad \alpha_k \sim \mathcal{N}(0, \sigma_\alpha^2 I_3), \quad (\text{A.7})$$

$$\eta_k^{bg} \sim \mathcal{N}(0, \sigma_{bg, \text{rw}}^2 I_3), \quad \eta_k^{ba} \sim \mathcal{N}(0, \sigma_{ba, \text{rw}}^2 I_3). \quad (\text{A.8})$$

The sampled state shown in the plots is the filter-style state

$$x_k = (R_k, p_k, v_k, b_{g,k}, b_{a,k}),$$

while the auxiliary variable ω_k is used internally only to generate the orientation process. For display we convert R_k into roll, pitch, and yaw angles. This is a visualization choice only: the actual filter should still represent rotational uncertainty in local Lie coordinates or in a quaternion plus 3D error state.

A.1 Prior values used for the illustrative samples

Quantity	Value	Role in the prior sampler
Time step Δt	1/60 s	Matches the camera-frame discretization used throughout the report.
Horizon T	4 s	Short lab-style motion segment.
σ_{p0}	(0.01, 0.01, 0.01) m	Initial position spread.
σ_{v0}	(0.03, 0.03, 0.03) m/s	Initial velocity spread.
$\sigma_{\text{rpy},0}$	(2, 2, 3) $^\circ$	Initial orientation spread, shown in Euler angles only for display.
$\sigma_{\omega 0}$	(4, 4, 5) $^\circ$ /s	Initial angular-velocity spread used to drive the pose process.
σ_a	0.10 m/s 2	Translational acceleration roughness.
σ_α	6 $^\circ$ /s 2	Angular-acceleration roughness.
$\sigma_{bg,0}$	(0.20, 0.20, 0.20) $^\circ$ /s	Initial gyroscope-bias spread.
$\sigma_{ba,0}$	(0.03, 0.03, 0.03) m/s 2	Initial accelerometer-bias spread.
$\sigma_{bg,\text{rw}}$	0.005 $^\circ$ /s per step	Gyroscope-bias random-walk step scale.
$\sigma_{ba,\text{rw}}$	0.001 m/s 2 per step	Accelerometer-bias random-walk step scale.

Table A.1: Illustrative prior values used for the annex sampler. These values are not claimed to be universal; they are chosen to produce motions in a realistic phone-on-arm laboratory regime, with translation excursions on the order of a few tenths of a meter and orientation excursions on the order of a few tens of degrees over the 4 s horizon.

The script used to generate the figures in this annex is stored alongside the report as `sample_prior_paths.py`, together with the JSON file `prior_sample_parameters.json`. The figures can therefore be regenerated whenever the assumed prior regime is adjusted.

The parameters in this appendix are discrete sampler settings. In particular, σ_a and σ_α are the standard deviations of the acceleration-like samples drawn at each simulation step, and $\sigma_{bg,\text{rw}}$ and $\sigma_{ba,\text{rw}}$ are per-step bias increments. They should not be read as the continuous-time spectral-density matrices $Q_v, Q_\omega, Q_{bg}, Q_{ba}$ without applying the appropriate Δt scaling.

A.2 Three-dimensional prior samples

Figure A.1 shows eight sampled translation paths. The curves begin near the origin because the initial position prior is tight, then fan out as the sampled accelerations accumulate. This picture is the translational part of the prior alone; it is *not* a posterior trajectory and it is not conditioned on data.

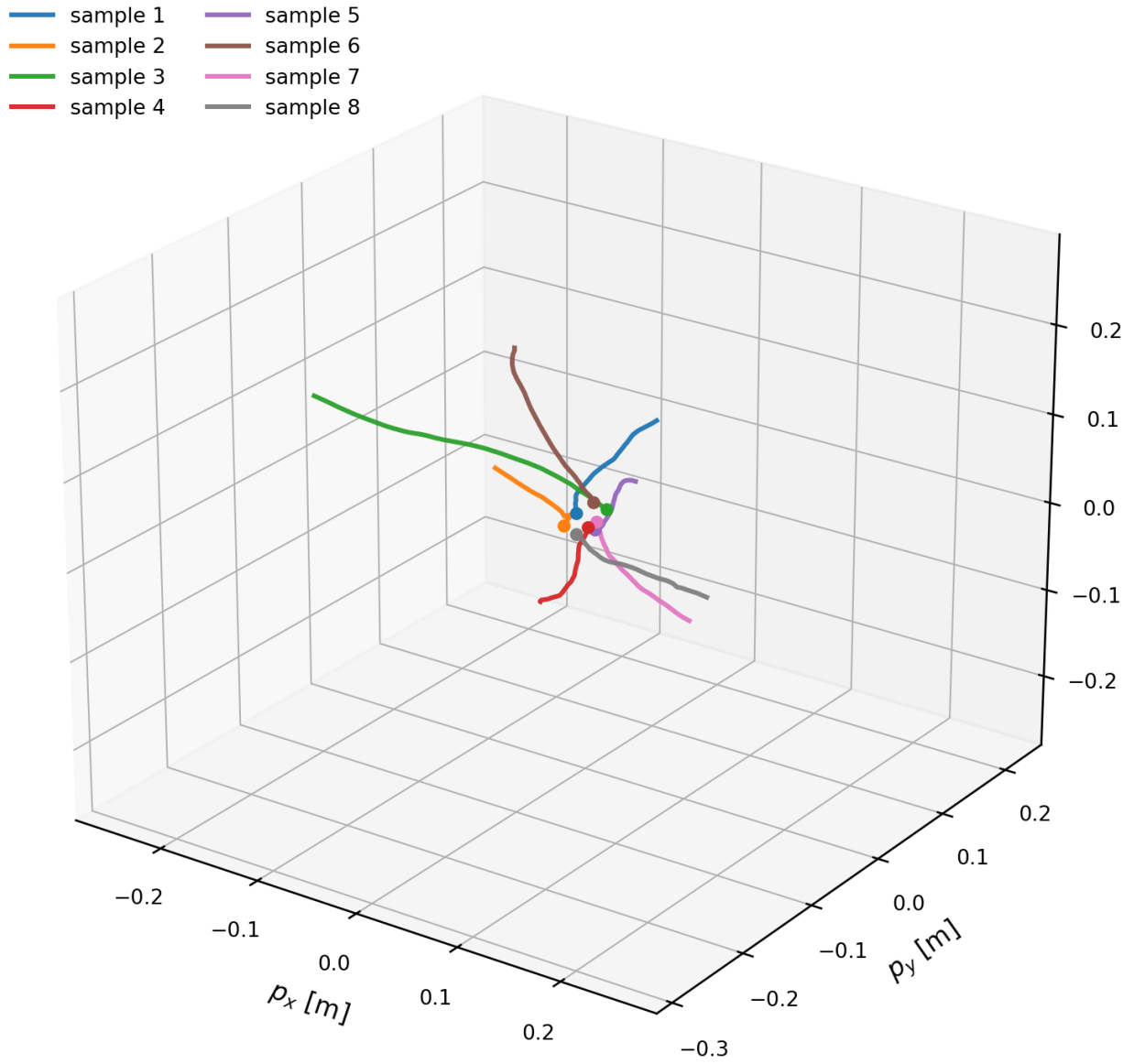


Figure A.1: Eight sample trajectories drawn from the discretized prior over a 4 s interval at 60 Hz. The paths illustrate the translational component of the prior before any measurement update is applied.

To visualize that the prior actually lives on poses rather than on positions only, Figure A.2 overlays body-frame triads along one sample path. The translational path lies in $\mathbb{R}^3$, but the orientation evolves on $SO(3)$ and is injected through the exponential-map update.

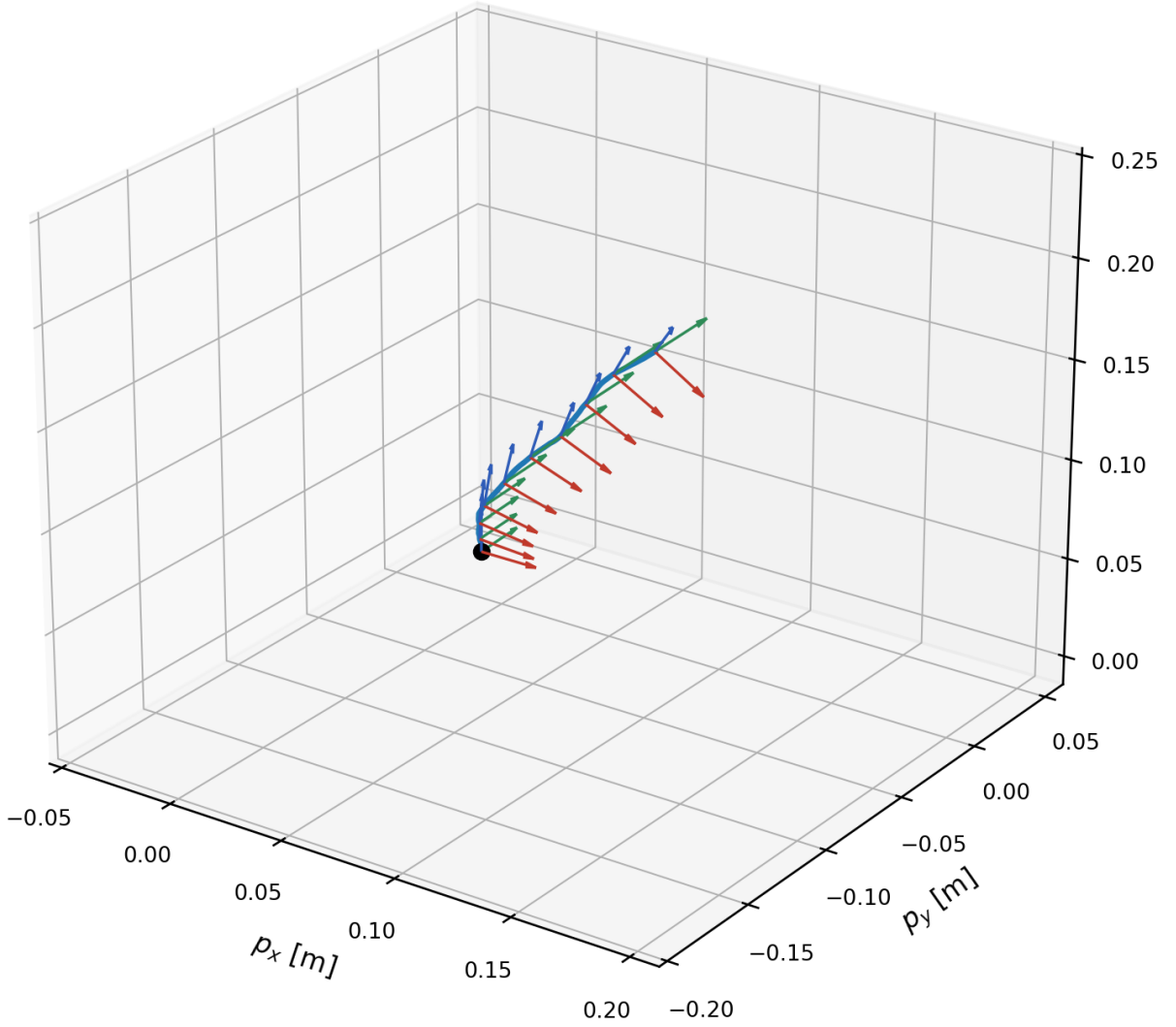


Figure A.2: One sampled pose trajectory with body-frame triads along the path. The translation is a path in $\mathbb{R}^3$, while the orientation component evolves on $SO(3)$ and is updated multiplicatively.

A.3 State evolution of all filter variables

Figure A.3 shows the evolution of all entries of the filter-style state

$$x_k = (R_k, p_k, v_k, b_{g,k}, b_{a,k}).$$

For readability, the orientation is displayed as roll, pitch, and yaw angles. The plots make the mixed geometry of the state visible: positions and velocities behave like ordinary Euclidean random walks with smoothness; the bias states evolve much more slowly; and the orientation display coordinates drift in a way that reflects accumulated angular velocity, even though the actual internal pose update is performed on the Lie group rather than by angle-wise addition.

A.4 How these samples should be interpreted

The figures in this annex are useful because they connect the abstract prior hyperparameters to visible trajectory behavior.

- Increasing σ_a makes the translational paths rougher and increases position spread rapidly over time.

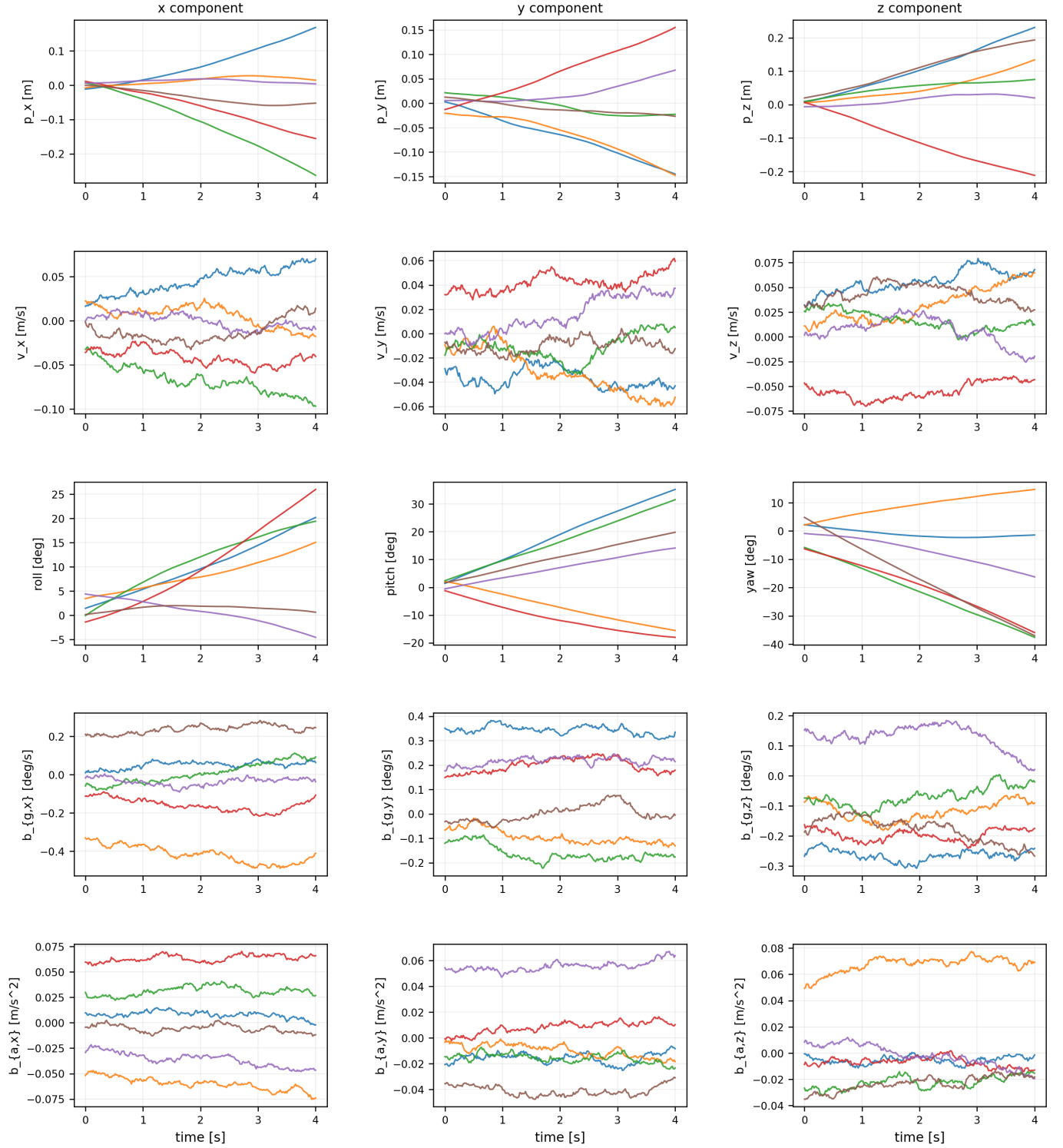


Figure A.3: Evolution of all filter-state entries for six prior samples. The third row displays orientation through roll, pitch, and yaw angles for visualization only. In the actual Bayesian formulation, the orientation posterior is represented locally in tangent coordinates or through a quaternion-plus-error representation rather than by filtering Euler angles directly.

- Increasing σ_α makes the orientation process wander more aggressively.
- Increasing $\sigma_{bg,rw}$ or $\sigma_{ba,rw}$ produces visibly drifting bias traces, which later appear in the filter as slower but persistent nuisance evolution.
- Tightening the initial spreads $\sigma_{p0}, \sigma_{v0}, \sigma_{rpy,0}$ shrinks the cloud near the start of the sequence, but it does not remove dynamic uncertainty later in time.

This is exactly why the prior deserves as much attention as the likelihood. Before any measurements are assimilated, the prior already says what kinds of motion are considered plausible. In a Kalman filter or IEKF, the same assumptions reappear as the initial covariance and the process-noise matrices. In a fixed-lag smoother or factor graph, they reappear as Gaussian motion factors. In a path-space Bayesian formulation, they are the law that generates the random curve itself.

Bibliography

- [1] Joan Solà, Jeremie Deray, and Dinesh Atchuthan. A micro Lie theory for state estimation in robotics, 2018. URL <https://arxiv.org/abs/1812.01537>.
- [2] Axel Barrau and Silvére Bonnabel. The invariant extended kalman filter as a stable observer. *IEEE Transactions on Automatic Control*, 62(4):1797–1812, 2017. doi: 10.1109/TAC.2016.2594085. URL <https://doi.org/10.1109/TAC.2016.2594085>.
- [3] Joan Solà. Quaternion kinematics for the error-state kalman filter, 2017. URL <https://arxiv.org/abs/1711.02508>.
- [4] Andrew M. Stuart. Inverse problems: A bayesian perspective. *Acta Numerica*, 19:451–559, 2010. doi: 10.1017/S0962492910000061. URL <https://doi.org/10.1017/S0962492910000061>.
- [5] Ajay Apte, Martin Hairer, Andrew M. Stuart, and Jochen Voss. Sampling the posterior: An approach to non-gaussian data assimilation. *Physica D: Nonlinear Phenomena*, 230(1–2):50–64, 2007. doi: 10.1016/j.physd.2006.06.009. URL <https://doi.org/10.1016/j.physd.2006.06.009>.
- [6] ETH Zurich Autonomous Systems Lab (ASL). Imu noise model. GitHub wiki, 2023. URL <https://github.com/ethz-asl/kalibr/wiki/IMU-Noise-Model>. Accessed 2026-04-25.
- [7] Anastasios I. Mourikis and Stergios I. Roumeliotis. A multi-state constraint kalman filter for vision-aided inertial navigation. In *Proceedings 2007 IEEE International Conference on Robotics and Automation*, pages 3565–3572. IEEE, April 2007. doi: 10.1109/ROBOT.2007.364024. URL <https://doi.org/10.1109/ROBOT.2007.364024>.
- [8] Christian Forster, Luca Carlone, Frank Dellaert, and Davide Scaramuzza. On-manifold preintegration for real-time visual-inertial odometry. *IEEE Transactions on Robotics*, 33(1):1–21, 2017. doi: 10.1109/TRO.2016.2597321. URL <https://doi.org/10.1109/TRO.2016.2597321>.
- [9] Manon Kok, Jeroen D. Hol, and Thomas B. Schön. Using inertial sensors for position and orientation estimation. *Foundations and Trends in Signal Processing*, 11(1–2):1–153, 2017. doi: 10.1561/20000000094. URL <https://doi.org/10.1561/20000000094>.
- [10] Edwin Olson. AprilTag: A robust and flexible visual fiducial system. In *2011 IEEE International Conference on Robotics and Automation*, pages 3400–3407. IEEE, May 2011. doi: 10.1109/ICRA.2011.5979561. URL <https://doi.org/10.1109/ICRA.2011.5979561>.
- [11] Syed Muhammad Abbas, Salman Aslam, Karsten Berns, and Abubakr Muhammad. Analysis and improvements in apriltag based state estimation. *Sensors*, 19(24):5480, 2019. doi: 10.3390/s19245480. URL <https://doi.org/10.3390/s19245480>.
- [12] Roman Adámek, Martin Brabc, Patrik Vávra, Barnabás Dobossy, Martin Formánek, and Filip Radil. Analytical models for pose estimate variance of planar fiducial markers for mobile robot localisation. *Sensors*, 23(12):5746, 2023. doi: 10.3390/s23125746. URL <https://www.mdpi.com/1424-8220/23/12/5746>.
- [13] Stefan Leutenegger, Simon Lynen, Michael Bosse, Roland Siegwart, and Paul Furgale. Keyframe-based visual-inertial odometry using nonlinear optimization. *The International Journal of Robotics Research*, 34(3):314–334, 2015. doi: 10.1177/0278364914554813. URL <https://doi.org/10.1177/0278364914554813>.
- [14] Michael Kaess, Hordur Johannsson, Richard Roberts, Viorela Ila, John J. Leonard, and Frank Dellaert. iSAM2: Incremental smoothing and mapping using the bayes tree. *The International Journal of Robotics Research*, 31(2):216–235, 2012. doi: 10.1177/0278364911430419. URL <https://doi.org/10.1177/0278364911430419>.
- [15] Simo Särkkä and Lennart Svensson. *Bayesian Filtering and Smoothing*. Cambridge University Press, 2 edition, 2023. ISBN 9781108926645. doi: 10.1017/9781108917407. URL <https://doi.org/10.1017/9781108917407>.
- [16] Google. Pixel 9a specs. Google Store, 2025. URL https://store.google.com/product/pixel_9a_specs. Accessed 2026-04-25.
- [17] Android Developers. Sensors overview. Android Developers documentation, 2026. URL https://developer.android.com/develop/sensors-and-location/sensors/sensors_overview. Accessed 2026-04-25.
- [18] Franka Robotics. *Datasheet Franka Research 3*. Franka Robotics, 2.4 edition, October 2025. URL https://franka.de/hubfs/Datasheet_FrankaResearch3_R02212_2.4.pdf?hsLang=en. Product page: <https://franka.de/franka-research-3>; FCI overview: <https://support.franka.de/docs/overview.html>; accessed 2026-04-25.

- [19] ETHZ ASL / kalibr wiki contributors. Yaml formats. GitHub wiki, 2018. URL <https://github.com/ethz-asl/kalibr/wiki/Yaml-formats/49be80e7695db848f6815d584d6c7cb28d3e390e>. Landing page: <https://github.com/ethz-asl/kalibr/wiki/Yaml-formats>; accessed 2026-04-25.
- [20] Bernd Pfrommer and Kostas Daniilidis. Tagslam: Robust slam with fiducial markers. *arXiv preprint arXiv:1910.00679*, 2019. doi: 10.48550/arXiv.1910.00679. URL <https://arxiv.org/abs/1910.00679>.
- [21] Christian Forster, Luca Carlone, Frank Dellaert, and Davide Scaramuzza. Imu preintegration on manifold for efficient visual-inertial maximum-a-posteriori estimation. In *Proceedings of Robotics: Science and Systems*, Rome, Italy, July 2015. doi: 10.15607/RSS.2015.XI.006. URL <https://www.roboticsproceedings.org/rss11/p06.html>.
- [22] Tong Qin, Peiliang Li, and Shaojie Shen. Vins-mono: A robust and versatile monocular visual-inertial state estimator. *IEEE Transactions on Robotics*, 34(4):1004–1020, 2018. doi: 10.1109/TRO.2018.2853729. URL <https://doi.org/10.1109/TRO.2018.2853729>.
- [23] Tim Barfoot, Chi Hay Tong, and Simo Särkkä. Batch continuous-time trajectory estimation as exactly sparse gaussian process regression. In *Proceedings of Robotics: Science and Systems*, volume 10, Berkeley, USA, July 2014. ISBN 9780992374709. doi: 10.15607/RSS.2014.X.001. URL <https://www.roboticsproceedings.org/rss10/p01.html>.
- [24] Sean Anderson, Timothy D. Barfoot, Chi Hay Tong, and Simo Särkkä. Batch nonlinear continuous-time trajectory estimation as exactly sparse gaussian process regression. *Autonomous Robots*, 39(3):221–238, 2015. doi: 10.1007/s10514-015-9455-y. URL <https://doi.org/10.1007/s10514-015-9455-y>.
- [25] M. Sanjeev Arulampalam, Simon Maskell, Neil Gordon, and Tim Clapp. A tutorial on particle filters for online nonlinear/non-gaussian bayesian tracking. *IEEE Transactions on Signal Processing*, 50(2):174–188, 2002. doi: 10.1109/78.978374. URL <https://doi.org/10.1109/78.978374>.